Beyond the Medical Model: The Two-Type Theory of Human Cognition and the Proposal for Cognitive Segregation (Voluntary Separation as Liberation)

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Abstract

The medical model of developmental disorders enumerates symptoms but fails to capture the lived experiences of individuals. This paper proposes a hypothesis that there exist two fundamentally distinct types of human cognition: hierarchical and non-hierarchical. In hierarchical cognition, information is organized in hierarchical structures, whereas in non-hierarchical cognition, it exists in network structures. This fundamental difference in cognitive structure expands from a four-quadrant model to an eight-quadrant model through combinations of dominant modality (language-dominant or body-dominant) and levels of metacognition (high, moderate, or low).

This framework enables the reinterpretation of existing diagnoses as cognitive structures. Autism Spectrum Disorder (ASD) represents the difficulty hierarchical cognition encounters when facing elements that cannot be hierarchically organized, manifesting differently across language-dominant and body-dominant types according to metacognitive levels. Attention-Deficit/Hyperactivity Disorder (ADHD) represents the difficulty non-hierarchical cognition faces in adapting to hierarchically structured societies, divided into language-dominant (predominantly inattentive) and body-dominant (predominantly hyperactive-impulsive) presentations. The co-occurrence of ASD and ADHD is explained uniformly as a state with moderate levels of metacognition.

Contemporary society is optimized for hierarchical cognition in language, institutions, and economy, forcing individuals with non-hierarchical cognition to bear the chronic burden of "translating" their cognitive style into hierarchical forms (translation cost). This translation cost manifests as physical symptoms, mental exhaustion, and social maladjustment. Rather than "inclusion," which functions as de facto assimilation pressure, this paper proposes "Structural Differentiation." Structural Differentiation refers to the deliberate design of social, institutional, and informational environments optimized for distinct cognitive structures, enabling each cognitive type to live in environments aligned with their inherent organization principles. This can be realized spatially, institutionally, and informationally, reducing translation costs and maximizing the capacities of each type.

This theory reconceptualizes developmental disorders not as "disorders" but as "differences in cognitive type," calling for the redesign of social structures rather than the treatment of individuals. Future directions include elucidating the neuroscientific foundations, developing measurement methods for the eight-quadrant model, and conducting experiments in Structural Differentiation.

1 Introduction

1.1 The Gap Between Lived Experience and Diagnostic Criteria

Individuals who have received a diagnosis of a developmental disorder—and those who have not been formally diagnosed but recognize that their cognitive style differs from what is considered "typical development"—often experience a profound gap between existing diagnostic criteria and their own lived experience. Contemporary diagnostic systems such as the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5; American Psychiatric Association, 2013) and the *International Classification of Diseases* (ICD-11; World Health Organization, 2019) define Autism Spectrum Disorder (ASD) and Attention-Deficit/Hyperactivity Disorder (ADHD) primarily through enumerations of observable behavioral symptoms.

ASD is characterized by "persistent deficits in social communication and social interaction" and "restricted, repetitive patterns of behavior, interests, or activities," while ADHD is described in terms of "inattention," "hyperactivity," and "impulsivity." These descriptions, however, are limited to externally observable behaviors. They do not adequately capture the internal cognitive processes through which individuals actually experience the world.

For example, what is labeled as "inattention" may reflect not a lack of attention, but a cognitive state in which *all information attracts attention equally.* Similarly, what is described as a "deficit in social communication" may arise from an inability to hierarchically organize implicit social rules, rather than from a lack of social motivation or concern. Diagnostic criteria tend to catalogue what individuals *cannot* do, but they do not explain *why* these difficulties arise or *how* the world is cognitively structured for the individual experiencing them.

An additional and more fundamental problem is that these diagnostic frameworks presuppose the concept of "disorder." Many individuals do not experience themselves as "malfunctioning versions of typical development." Rather, they live with a persistent sense of friction—an ongoing question of why everyday life feels disproportionately exhausting, confusing, or overwhelming. While this friction is often medicalized as a collection of symptoms, its deeper source may lie in a mismatch between a person's cognitive structure and a society optimized for a different cognitive type.

The author's own long-term self-analysis emerged from precisely this unresolved sense of mismatch. Through systematic reflection on personal cognitive processes, this analysis converged on an interpretation based not on deficit, but on the existence of fundamentally different cognitive types. This paper seeks to formalize that interpretation and to provide conceptual language and structure for others who experience a similar form of cognitive dissonance.

1.2 Limitations of Existing Approaches

Several influential approaches have shaped contemporary understandings of developmental disorders, yet each remains limited in its ability to account for fundamental differences in cognitive structure.

The **medical model**, as operationalized in DSM-5 and ICD-11, has provided standardized diagnostic categories and access to support. However, its focus remains on surface-level behavioral manifestations rather than on the internal organization of cognition. Moreover, the medical model frames "adaptation" to existing social structures as the primary therapeutic goal, implicitly requiring individuals to conform to a hierarchically structured society. If much of the experienced suffering originates not from intrinsic dysfunction but from a mismatch between cognitive type and social structure, this model addresses symptoms while leaving the underlying problem intact.

The **social model of disability** (Oliver, 1990) shifts attention from individual impairment to socially constructed barriers and advocates inclusion through environmental accommodation. While this perspective represents a major ethical and political advance, it encounters limitations when applied to non-hierarchical cognition. In such cases, barriers are not merely external obstacles but are embedded in the very structure of language, reasoning, and communication within a hierarchically organized society. Standard forms of "reasonable accommodation" often presuppose the existing structure and thus generate ongoing trade-offs that function, in practice, as assimilation pressure rather than genuine inclusion.

The **neurodiversity movement** (Singer, 1998) challenges the pathologization of developmental differences by framing them as part of natural human variation. However, while it emphasizes acceptance and respect, it has not yet produced a systematic theoretical account of *how* cognitive structures differ, *why* these differences arise, or *why* coexistence within a single social structure is often so difficult. The concept of diversity alone does not explain the mechanisms that generate persistent cognitive friction.

Taken together, these approaches have made essential contributions, yet none offers a unified theory that explains developmental diagnoses in terms of fundamental cognitive structures or derives concrete implications for social design from those differences. This paper aims to address that gap.

1.3 Aims and Structure of the Present Paper

The purpose of this paper is to reconceptualize developmental disorders as expressions of fundamental differences in cognitive structure and to explore the social implications of those differences. Specifically, this paper pursues three main objectives.

First, it proposes a theoretical framework in which human cognition is divided into two fundamentally different types: **hierarchical cognition** and **non-hierarchical cognition**. In hierarchical cognition, information is organized into tree-like structures characterized by inclusion, ordering, and prioritization. In non-hierarchical cognition, information exists as a network or graph, organized through parallel, associative connections. This basic distinction is further elaborated through the introduction of **dominant modality** (language-dominant vs. body-dominant) and **levels of metacognition** (high, moderate, low), expanding an initial four-quadrant model into an eight-quadrant model. This framework allows existing diagnoses such as ASD, ADHD, and AuDHD to be reinterpreted as positions within a multidimensional cognitive space rather than as discrete disorders.

Second, the paper analyzes the structural asymmetry produced by the fact that contemporary society—across language, institutions, and economic systems—is optimized for hierarchical cognition. Individuals with non-hierarchical cognition are therefore compelled to translate their cognitive processes into hierarchical forms, incurring a chronic cognitive burden referred to here as **translation cost**. The manifestation of this cost varies depending on dominant modality and level of metacognition, appearing as physical symptoms, psychological exhaustion, or social maladaptation.

Third, as a response to this structural asymmetry, the paper proposes **Structural Differentiation** as an alternative to models centered on adaptation or coexistence. Structural Differentiation refers to the possibility that different cognitive types live within environments optimized for their own cognitive structures—spatially, institutionally, and informationally—thereby reducing translation cost and enabling each type's capacities to be fully realized.

The paper proceeds as follows. Chapter 2 introduces the theoretical framework of hierarchical and non-hierarchical cognition, first through the four-quadrant model and then through its extension into the eight-quadrant model incorporating metacognition. Chapter 3 reinterprets existing neurodevelopmental diagnoses within this framework, with particular attention to differences in modality and developmental trajectories across metacognitive levels. Chapter 4 analyzes the concept of translation cost and the structural asymmetry of hierarchical societies. Chapter 5 develops Structural Differentiation as a practical proposal for social design. Chapter 6 discusses the limitations, falsifiability, and future research directions of the theory, including the need for measurement methods and empirical validation of each dimension of the model.

The theoretical development presented here is grounded primarily in long-term self-analysis by the author, a form of autoethnographic inquiry with the inherent limitation of an $n = 1$ sample. The author is situated within the eight-quadrant model as a **non-hierarchical** $\times$ **language-dominant** $\times$ **high-metacognition** type. This positionality may introduce bias into interpretations of other quadrants. Nevertheless, detailed articulation of internal cognitive experience can illuminate structural features of cognition that remain difficult to capture through quantitative methods alone. Future research should therefore include multiple case studies across quadrants, empirical testing of the model, and investigation of its potential neurobiological correlates.

2 Theoretical Framework

In response to the gap between diagnostic criteria for developmental disorders and lived experience discussed in the previous chapter, this chapter presents a new theoretical framework for understanding human cognitive structures.

2.1 The Two-Type Theory of Human Cognition: Hierarchical and Non-Hierarchical Types

The present theory distinguishes human cognitive structures into two fundamentally different types. This distinction is based on the **principle of information organization**, that is, how cognitive systems organize, compress, and handle information.

2.1.1 Hierarchical Cognition

Information organization principle: hierarchy, tree structure

Individuals with hierarchical cognition organize information according to **logical inclusion relations, sequential order, and exclusivity**.

- **Feature A: Linearity** Thinking proceeds in a linear manner, allowing explanations to move systematically from conclusions to reasons, or from higher-level categories to lower-level elements.
- **Feature B: Inclusion** Concepts are classified through clearly defined superordinate and subordinate categories, and information is positioned within a coherent tree-like structure.
- **Feature C: Prioritization** Hierarchical cognition excels at ranking information by importance or urgency and determining an appropriate order of processing.
- **Compatibility with society** Most modern social systems—including bureaucracy, educational curricula, legal systems, and classical logic—are optimized for hierarchical cognition.

2.1.2 Non-Hierarchical Cognition

Information organization principle: network, graph, galaxy-like structure

Individuals with non-hierarchical cognition organize information through **parallel, simultaneous, and associative links** rather than through inclusion-based hierarchies.

- **Feature A: Parallel processing** Multiple pieces of information or tasks are processed simultaneously. Information tends to carry equal weight and coexist concurrently in conscious awareness.
- **Feature B: Associative linking** Concepts are connected not by hierarchical inclusion but by contextual, sensory, or temporal proximity. Thought is non-linear and can jump rapidly across multiple nodes.
- **Feature C: Fluidity and responsiveness** Because structures are not fixed, non-hierarchical cognition is highly responsive to environmental change and well suited to creative insight arising from unclassified information.
- **Mismatch with society** Hierarchical institutions and linear communication practices (e.g., turn-taking in meetings, ordered reporting) impose excessive cognitive burden—hereafter referred to as **translation cost**—on individuals with non-hierarchical cognition.

2.1.3 Cognitive Structure as Information Compression: Discretization and Continuity

The difference between hierarchical and non-hierarchical cognition can be understood as a difference in **information compression strategies**.

Hierarchical cognition transforms continuous real-world input into discrete symbols such as language and categories. This process resembles irreversible, lossy compression: some information is discarded in exchange for ease of organization, transmission, and manipulation. Major social institutions—law, education, and formalized procedures—are optimized for this strategy of discretization.

By contrast, non-hierarchical cognition tends to preserve continuous information such as sensory input, context, and simultaneity as faithfully as possible, exhibiting a stronger orientation toward lossless representation. While this allows the retention of high-dimensional information, converting such representations into discrete formats requires substantial cognitive effort.

There exists an asymmetry in translation cost between these two strategies. Discretized information can be readily embedded within continuous representations, whereas translating continuous representations into discrete structures necessarily involves information loss. This asymmetry provides a structural explanation for why individuals with non-hierarchical cognition experience heightened explanatory burden and stress associated with verbalization.

2.2 From Four Quadrants to Eight: A Model of Cognitive Differentiation

This section refines the basic distinction between hierarchical and non-hierarchical cognition by introducing additional dimensions. First, a four-quadrant model incorporating dominant modality is presented. This is then expanded into an eight-quadrant model by incorporating level of metacognition.

2.2.1 The Four-Quadrant Model: Cognitive Structure × Dominant Modality

Human cognition is shaped not only by information organization (hierarchical vs. non-hierarchical) but also by **dominant modality**, the mode through which individuals preferentially perceive, think, and express.

Language-dominant cognition processes the world primarily through language, logic, and abstract concepts. Thinking is largely verbalized, and understanding deepens through linguistic explanation. Domains such as mathematics, philosophy, literature, and law tend to reward this modality.

Body-dominant cognition processes the world primarily through bodily sensation, spatial perception, movement, and visual imagery. Thinking is closely tied to physical experience, and understanding develops through practice and embodied interaction. Domains such as sports, art, craftsmanship, and architecture tend to reward this modality.

Combining these two dimensions yields four basic quadrants:

	Language-dominant	**Body-dominant**
Hierarchical	Logical–linguistic systemizers	Craft-based or motor systemizers
Non-hierarchical	Creative linguistic explorers	Artistic or embodied explorers

Individuals with **hierarchical × language-dominant** cognition systematically organize the world through language and logic. They are common among scholars, legal professionals, and bureaucrats. However, they may struggle with tacit knowledge, nonverbal communication, and subtle emotional cues.

Individuals with **hierarchical × body-dominant** cognition organize the world hierarchically through space and action. They are common among artisans, engineers, and architects. While abstract verbal explanation may be difficult, they excel in procedural knowledge acquisition and precise bodily control.

Individuals with **non-hierarchical × language-dominant** cognition associate concepts in network-like ways, producing rapid conceptual leaps. They are common among creative scientists, writers, and strategists. Linear social demands—such as strict time management or stepwise explanation—often incur substantial translation cost.

Individuals with **non-hierarchical × body-dominant** cognition explore space and sensation in a network-like manner and respond improvisationally. They are common among artists, dancers, and designers, and may resist verbalization or rigid planning.

2.2.2 Expansion via Metacognition: The Eight-Quadrant Model

While the four-quadrant model captures fundamental combinations of cognitive structure and modality, substantial individual differences remain within each quadrant. A critical dimension accounting for this variation is **level of metacognition**.

Metacognition refers to the ability to recognize, monitor, and regulate one's own cognitive processes. Individuals with high metacognition can understand their cognitive tendencies and adjust them according to situational demands. Those with low metacognition are often unaware of their cognitive style and experience difficulty adapting when mismatches arise.

By dividing metacognition into three levels (high, moderate, low), the four quadrants expand into eight clinically salient positions (strictly speaking, twelve are possible, but eight are most relevant):

Table 1: Table caption

Cognitive structure	Dominant modality	High metacognition	Moderate metacognition	Low metacognition
Hierarchical	**Language-dominant**	Logical scholars, legal professionals (typically developed)	Linguistic systemizers with weak tacit knowledge (mild ASD)	Linguistic disorganization: severe difficulty with social context (severe ASD)
Hierarchical	**Body-dominant**	Skilled artisans, engineers, architects (typically developed)	Motor systemizers with weak abstraction (mild ASD)	Bodily disorganization: severe difficulty with abstract language (severe ASD)
Non-hierarchical	**Language-dominant**	Creative scientists, strategists ("galaxy type")	Linguistic explorers burdened by translation (AuDHD)	Linguistic diffusion: attentional overload (ADHD, inattentive type)
Non-hierarchical	**Body-dominant**	Artists, dancers, designers ("galaxy type")	Embodied explorers with social friction (AuDHD)	Bodily impulsivity: impaired behavioral control (ADHD, hyperactive type)

This eight-quadrant model demonstrates that existing diagnostic categories—ASD, ADHD, and AuDHD—can be understood as positions within a three-dimensional space defined by cognitive structure, dominant modality, and level of metacognition. The following sections examine each quadrant in detail.

2.2.3 Three Levels of Hierarchical × Language-Dominant Cognition

High Metacognition: Logical Scholars and Legal Professionals (Typically Developed)

Individuals at this level possess a strong hierarchical and language-dominant cognitive structure and are able to consciously understand, monitor, and regulate their own cognitive processes. They excel in logical reasoning, abstract conceptualization, and the systematic construction of knowledge. Implicit social rules can be acquired through observation and explicit learning, allowing relatively smooth participation in social contexts.

Because their cognitive structure aligns closely with the dominant organization of contemporary society, they are generally regarded as "typically developed." They tend to thrive in fields that demand linguistic and logical systematization, such as academia, law, management, and bureaucracy.

Moderate Metacognition: Linguistic Systemizers with Weakness in Tacit Knowledge (Mild ASD)

Individuals at this level demonstrate exceptional abilities in logical and linguistic systematization, yet their metacognitive capacity is only moderate. As a result, they struggle with domains dominated by tacit knowledge, such as nonverbal communication, emotional nuance, and fluid social contexts.

They possess partial awareness of their own cognitive style but have limited capacity to explicate or flexibly apply implicit social rules. Consequently, interpersonal friction often arises in everyday communication. However, within areas of strong interest, they may achieve an extraordinary degree of expertise by refining hierarchical structures to a high level of precision.

Many individuals previously described as having "high-functioning autism" or Asperger's syndrome fall within this category.

Low Metacognition: Linguistic Chaos and Severe Difficulty with Social Contexts (Severe ASD)

At this level, hierarchical tendencies in cognition remain present, but low metacognition prevents effective self-regulation. Logical coherence is difficult to maintain, and understanding social context is profoundly impaired. Even when linguistic ability itself is preserved, it cannot be effectively applied to social communication.

Typical manifestations include echolalia, scripted speech, one-sided monologues, heightened sensory hypersensitivity, intense fixations, and panic responses to unpredictability. Individuals in this category often require substantial support. Because verbal fluency may mask underlying difficulties, their challenges are frequently misunderstood or underestimated.

2.2.4 Three Levels of Hierarchical × Body-Dominant Cognition

High Metacognition: Craft-Oriented Engineers and Architects (Typically Developed)

Individuals at this level excel in spatial cognition, bodily control, and the hierarchical organization of procedural knowledge. They are able to reflect on their bodily and spatial cognition and, when necessary, translate it into linguistic form.

They perform well in domains that require precise bodily control and spatial reasoning, such as craftsmanship, engineering, architecture, and surgery. While verbal explanation may not be their primary strength, they can systematically refine skills through practice and embodied experience.

Moderate Metacognition: Motor Systemizers with Weakness in Abstraction (Mild ASD)

These individuals are adept at organizing knowledge through bodily movement and spatial patterns but experience difficulty with abstract language, logical verbal explanation, and socially mediated linguistic communication.

They possess partial awareness of their bodily cognitive style but have limited ability to verbalize it. As a result, they often struggle in language-centered educational environments while demonstrating exceptional competence in practical or hands-on tasks. A subset of individuals traditionally described as having "classic autism" belongs to this category.

Low Metacognition: Bodily Chaos and Severe Difficulty with Abstract Language (Severe ASD)

At this level, bodily and spatial processing dominates cognition, but low metacognition prevents effective regulation. Linguistic communication is extremely limited or absent, and access to abstract reasoning is minimal.

Common manifestations include nonverbal or highly restricted speech, immersion in sensory experiences, repetitive motor behaviors, and self-stimulatory actions (stimming) used for sensory regulation. Individuals in this group typically require intensive support, as bodily cognition is overwhelmed by unstructured sensory input and cannot be integrated into the linguistic social world.

2.2.5 Three Levels of Non-Hierarchical × Language-Dominant Cognition

High Metacognition: Creative Scientists and Strategists (Galaxy Type)

Individuals with non-hierarchical and language-dominant cognition combined with high metacognition are capable of constructing a functional interface with hierarchically structured society. Through metacognitive awareness, they can consciously deploy hierarchical language and logic when required, while retaining non-linear and associative thinking.

They possess a deep understanding of their own cognitive structure and can flexibly shift between cognitive modes. Such individuals often excel in fields that require both non-linear creativity and formal explanation, including theoretical science, strategy, entrepreneurship, and interdisciplinary innovation. The author of this paper belongs to this category.

Moderate Metacognition: Linguistic Explorers Exhausted by Translation (AuDHD)

Individuals in this category possess non-hierarchical cognition and partial metacognitive awareness. They attempt to construct a translation layer to adapt to hierarchical social demands, but this layer remains incomplete.

Although they may appear socially functional, the continuous effort required for translation produces chronic exhaustion. Many individuals diagnosed with co-occurring ASD and ADHD (AuDHD), particularly those with language-dominant profiles, fall within this category.

Low Metacognition: Linguistic Diffusion and Loss of Focus (ADHD, Inattentive Presentation)

At this level, non-hierarchical parallel processing dominates cognition, and low metacognition results in all information attracting equal attention. Linguistic thought is active but persistently diffusive, making sustained focus on a single task difficult.

Hierarchical demands such as prioritization, planning, and time management are structurally incompatible with this cognitive organization. Consequently, individuals in this category are commonly evaluated as exhibiting "inattention," corresponding to the inattentive presentation of ADHD.

2.2.6 Three Levels of Non-Hierarchical × Body-Dominant Cognition

High Metacognition: Artists, Dancers, and Designers (Galaxy Type)

Individuals in this category combine non-hierarchical bodily cognition with high metacognition. They understand their embodied, improvisational cognition and can consciously respond to hierarchical social demands when necessary.

They excel in domains that require bodily improvisation and expressive creativity, such as visual arts, dance, design, and improvisational performance.

Moderate Metacognition: Bodily Explorers Experiencing Friction with Social Norms (AuDHD)

These individuals possess non-hierarchical bodily cognition and partial metacognitive awareness. They attempt to conform to social norms, but adaptation remains incomplete.

They often experience chronic tension between bodily impulsivity and social expectations, leading to sustained fatigue. Many individuals diagnosed with AuDHD with body-dominant profiles fall into this category.

Low Metacognition: Bodily Impulsivity and Impaired Behavioral Control (ADHD, Hyperactive-Impulsive Presentation)

At this level, non-hierarchical bodily processing predominates, and low metacognition prevents regulation of bodily impulses. Individuals are unable to remain still and tend to act immediately.

Hierarchical social requirements such as waiting, remaining stationary, and engaging in planned behavior are structurally incompatible with this cognitive organization. As a result, such individuals are commonly evaluated as exhibiting hyperactivity and impulsivity, corresponding to the hyperactive-impulsive presentation of ADHD.

2.3 Correspondence Between the Eight-Quadrant Model and Diagnostic Categories

The eight-quadrant model provides a foundation for reconceptualizing existing diagnostic categories not as "disorders," but as specific positions within a three-dimensional space defined by cognitive structure, dominant modality, and level of metacognition.

2.3.1 Positioning of Autism Spectrum Disorder (ASD)

Within this framework, ASD corresponds to quadrants characterized by **hierarchical cognition (language-dominant or body-dominant) combined with moderate to low metacognition**. The core difficulty arises when hierarchical cognition encounters elements that cannot be hierarchically organized, such as implicit rules, fluid relationships, and raw sensory input.

Differences in metacognitive level account for observed variations commonly described as "mild" versus "severe" or "high-functioning" versus "low-functioning," while differences in dominant modality explain linguistic versus bodily presentations.

2.3.2 Positioning of Attention-Deficit/Hyperactivity Disorder (ADHD)

ADHD corresponds to quadrants characterized by **non-hierarchical cognition combined with low metacognition**. Core difficulties emerge when non-hierarchical cognitive organization confronts the linear demands of hierarchically structured society, including planning, prioritization, and time management.

Differences between inattentive and hyperactive-impulsive presentations are explained as differences in dominant modality (language-dominant versus body-dominant), rather than as distinct underlying mechanisms.

2.3.3 Positioning of AuDHD

Within this framework, AuDHD is understood as a state characterized by **moderate metacognition**, regardless of whether the underlying cognitive structure is hierarchical or non-hierarchical.

At this level, individuals possess partial awareness of their cognitive style but lack sufficient metacognitive capacity to fully resolve the mismatch between their cognition and social demands. This results in chronic translation cost and characteristic exhaustion.

Two developmental origins of AuDHD can be distinguished:

1. A non-hierarchical origin, in which individuals attempt to construct an incomplete translation layer toward hierarchical norms.
2. A hierarchical origin, in which individuals attempt to incompletely acquire the processing of tacit knowledge.

2.3.4 An Integrative Perspective

This theoretical framework offers an integrative perspective in which ASD, ADHD, and AuDHD are understood as different positions within a shared cognitive space rather than as mutually contradictory diagnoses. Diagnostic labels function merely as convenient markers indicating approximate locations within this space.

What matters most is not the label itself, but the interaction between an individual's cognitive structure and the surrounding social environment—and how this interaction shapes lived experience.

3 Reinterpretation of Neurodevelopmental Diagnoses

Using the theoretical framework of hierarchical and non-hierarchical cognition proposed in the previous chapter, this chapter reinterprets existing neurodevelopmental diagnoses. Through this lens, the cognitive structures underlying what are labeled as "symptoms" by diagnostic categories are brought into clearer focus.

3.1 Reinterpretation of Autism Spectrum Disorder (ASD)

3.1.1 Conventional Understanding

In the DSM-5, Autism Spectrum Disorder (ASD) is defined by two core features: (1) persistent deficits in social communication and social interaction, and (2) restricted, repetitive patterns of behavior, interests, or activities (American Psychiatric Association, 2013).

Within cognitive psychology, one influential account is Baron-Cohen's systemizing theory, which characterizes ASD as a cognitive profile marked by a strong bias toward systemizing—understood as the drive to analyze, construct, and predict rule-based systems—accompanied by relatively weaker empathizing capacities. This approach has contributed to reframing ASD not merely as a deficit, but as a distinctive cognitive style.

However, these conventional accounts remain largely descriptive. They catalog observable behaviors and tendencies but do not provide a sufficiently explicit model of the underlying cognitive structure that generates these patterns, nor do they adequately explain why such traits become disabling in particular social contexts.

3.1.2 Reinterpretation within the Present Theory

Within the present theoretical framework, the core of ASD is reinterpreted as **the difficulty experienced by individuals with hierarchical cognition when they confront elements that cannot be hierarchically organized**.

Hierarchical cognition operates by organizing information into explicit structures defined by inclusion relations, priority, and stable categories. Social interaction, however, is saturated with non-hierarchical elements:

implicit rules, fluid relationships, context-dependent meanings, and rapidly shifting emotional cues. These elements resist stable hierarchical classification.

From this perspective, difficulties in social communication do not stem from a lack of social motivation or intelligence, but from a structural mismatch. Individuals with hierarchical cognition attempt to process social information by imposing explicit hierarchical organization, yet much of social reality is fundamentally non-hierarchical. This mismatch renders social interaction structurally difficult rather than merely effortful.

Restricted and repetitive patterns of behavior can be understood as expressions of a strong orientation toward structural stability and consistency. Routines, rituals, insistence on sameness, and narrow interests function as mechanisms for preserving hierarchical order in an otherwise unstable and unpredictable environment. Intense focus on specific domains reflects the hierarchical tendency to refine and elaborate a limited structure to a high degree of internal coherence.

Sensory hypersensitivity can likewise be reinterpreted as overload from unstructured sensory input. Hierarchical cognition seeks to discretize and organize sensory information, yet raw sensory experience is continuous and non-hierarchical by nature. When sensory input cannot be effectively hierarchized, excessive cognitive burden arises, resulting in hypersensitivity and distress.

3.1.3 Variability of Expression by Modality and Level of Metacognition

The manifestation of ASD varies substantially depending on the dominant modality (language-dominant vs. body-dominant) and the level of metacognition. This multidimensional variability explains phenomena that have traditionally been described using dichotomies such as "high-functioning vs. low-functioning" or "Asperger's syndrome vs. classical autism."

3.1.3.1 Language-Dominant ASD: Three Levels High Metacognition (Near-Typical Development): Logical Scholars and Legal Thinkers

Individuals in this category exhibit strong hierarchical organization through language and logic, combined with relatively high metacognitive awareness. As a result, they are able to compensate for difficulties with implicit social rules by explicitly learning and applying them. Social norms are treated as formal systems that can be studied, memorized, and consciously implemented.

At this level, ASD-related traits are often perceived as individual differences rather than impairments, and diagnostic thresholds may not be reached. Such individuals frequently thrive in domains that reward logical rigor and explicit reasoning, including academia, law, information technology, and scientific research.

Moderate Metacognition (High-Functioning ASD / Asperger's): Linguistic Systemizers with Weakness in Tacit Knowledge

These individuals demonstrate exceptional ability in linguistic and logical systemization, but their metacognitive capacity is insufficient to fully compensate for deficits in processing tacit knowledge. They partially understand their own cognitive style but struggle to make implicit social information explicit.

Typical difficulties include:

- Literal interpretation of metaphors and irony

- Difficulty timing conversational turn-taking
- Limited ability to infer others' emotional states
- Confusion between social conventions and sincere intentions

Despite these challenges, they often develop extraordinarily refined hierarchical structures within their domains of interest, achieving high levels of expertise and internal coherence. Many individuals historically labeled as having "high-functioning autism" or "Asperger's syndrome" fall into this category.

Low Metacognition (Severe ASD, Language-Dominant): Linguistic Chaos and Social Context Failure

At this level, hierarchical tendencies are present but metacognitive control is minimal. Individuals are unable to regulate their own cognitive processes or maintain logical coherence in social contexts. Although formal language ability may be preserved—sometimes including fluent speech and advanced vocabulary—it cannot be effectively deployed for social communication.

Typical manifestations include:

- Echolalia (repetition of others' utterances)
- Scripted or memorized conversational patterns
- One-sided monologues that ignore interlocutor feedback
- Extreme sensory hypersensitivity
- Intense insistence on sameness and panic responses to change

These individuals often require substantial support. Because surface-level language ability may appear intact, their difficulties are frequently misunderstood or underestimated.

3.1.3.2 Body-Dominant ASD: Three Levels High Metacognition (Near-Typical Development): Artisanal Experts and Architects

These individuals excel in hierarchical organization of bodily movement, spatial cognition, and procedural knowledge. With sufficient metacognitive awareness, they can intentionally translate bodily understanding into language when required. Although verbal explanation may be challenging, they achieve high levels of mastery through practice and embodied learning.

They are commonly found in fields that demand precise bodily control and spatial reasoning, such as craftsmanship, engineering, architecture, and surgery.

Moderate Metacognition (Classical ASD): Motor Systemizers with Weakness in Abstraction

Here, bodily and spatial systemization is strong, but abstraction and linguistic explanation are difficult. Individuals partially recognize their bodily cognitive style but lack the capacity to fully articulate it.

Typical difficulties include:

- Understanding verbal instructions that rely on vague or abstract language
- Comprehension of abstract concepts such as justice or friendship
- Verbal expression of internal states

- Academic environments centered on language-based instruction

Nevertheless, these individuals often demonstrate remarkable abilities in hands-on tasks, technical replication, and intuitive spatial understanding. Many individuals historically described as having "classical autism" belong to this group.

Low Metacognition (Severe ASD, Non-Language-Dominant): Bodily Chaos and Disconnection from Abstract Language

At this level, bodily and sensory processing dominates, but metacognitive regulation is minimal. Verbal communication is extremely limited or absent, and connection to abstract linguistic systems is weak.

Typical manifestations include:

- Nonverbal or minimally verbal communication
- Immersion in sensory experience
- Repetitive bodily movements (stereotypies)
- Self-stimulatory behaviors (stimming) as sensory regulation
- Extreme difficulty in reciprocal interaction

These individuals require extensive support. Their bodies are overwhelmed by sensory input, and linguistic access to the social world is severely constrained.

3.1.3.3 Historical Shifts in Diagnosis and the Role of Modality Historically, diagnostic frameworks distinguished between language-dominant presentations (e.g., Asperger's syndrome) and body-dominant presentations (e.g., classical autism). In DSM-IV, these were treated as separate diagnoses, while DSM-5 unified them under the single category of Autism Spectrum Disorder.

However, this unification obscured the internal diversity of ASD. The present theory clarifies that this diversity is not arbitrary but systematically structured by the interaction of cognitive structure, dominant modality, and level of metacognition. ASD is therefore not a single disorder but a constellation of positions within a three-dimensional cognitive space.

3.2 Reinterpretation of ADHD

3.2.1 Conventional Understanding

In DSM-5, Attention-Deficit/Hyperactivity Disorder (ADHD) is defined by two core symptom clusters: **inattention** and **hyperactivity–impulsivity**. Neuropsychological research has traditionally conceptualized ADHD as a disorder of **executive function** (Barkley, 1997). Executive functions refer to higher-order cognitive processes such as behavioral inhibition, working memory, planning, and problem solving.

However, while this framework describes *which functions fail to operate as expected* within a hierarchically structured society, it does not sufficiently explain *why* these functions fail, nor does it clarify the underlying **cognitive structure** that gives rise to ADHD-related behaviors. As a result, executive dysfunction is often treated as a deficit intrinsic to the individual, rather than as a mismatch between cognitive structure and social demands.

3.2.2 ADHD in the Present Theory: Non-Hierarchical Cognition and Dominant Modality

In the present theoretical framework, ADHD is reconceptualized as **the difficulty experienced by individuals with non-hierarchical cognition when adapting to a hierarchically structured society**—particularly one that demands planning, prioritization, linear time management, and rule-based behavioral inhibition.

Furthermore, differences in **dominant modality (language-dominant vs. body-dominant)** account for the traditional distinction between the inattentive and hyperactive–impulsive presentations of ADHD.

3.2.2.1 Non-Hierarchical × Language-Dominant × Low Metacognition: Predominantly Inattentive Type Cognitive Structural Characteristics

In this configuration, non-hierarchical parallel processing is dominant, and metacognition is low. As a result, **all incoming information attracts attention equally**, making selective focus structurally difficult. Linguistic and conceptual thinking is active and generative, but attention continuously shifts across the network of associations rather than converging on a single focal node.

The hierarchical operations required by modern society—such as distinguishing between important and unimportant tasks, sequencing actions, and sustaining attention over time—are therefore not automatically generated by the cognitive system.

The True Nature of "Inattention"

What is conventionally labeled as *inattention* is more accurately described as a state of **attending to everything simultaneously**. In non-hierarchical cognition, information exists within a flat network rather than a prioritized hierarchy. Consequently, the selective attentional filtering that hierarchical cognition performs automatically does not occur.

Typical manifestations include:

- **Shifting focus mid-task**: associative links lead attention from one node to another before task completion
- **Frequent loss or misplacement of objects**: spatial information is not hierarchically organized into stable location-based representations
- **Difficulty following multi-step instructions**: instructions exist in parallel rather than being retained as ordered sequences
- **Time management difficulties**: time cannot be organized into hierarchical structures such as deadlines, backward planning, and prioritization
- **Planning difficulties**: hierarchical stepwise representations ("first, then, finally") are unstable or absent

Expression of Language Dominance

Linguistic and conceptual activity is highly active. Ideas emerge rapidly and in abundance, but cannot be hierarchically organized into executable plans. In conversation, topics frequently jump through associative links, resulting in digressions and loss of the original thread. During reading, associative expansion may pull attention away from the primary textual context.

Social Evaluation

Individuals in this quadrant are often described as "intelligent but inexplicably incapable of execution." Despite high intellectual potential, they fail to meet the executive demands of hierarchical society and are frequently misinterpreted as lazy or unmotivated.

3.2.2.2 Non-Hierarchical × Body-Dominant × Low Metacognition: Predominantly Hyperactive–Impulsive Type Cognitive Structural Characteristics

In this configuration, non-hierarchical bodily processing is dominant, and metacognition is low. The body continuously explores the network of environmental affordances. Remaining physically still is equivalent to forcibly suppressing the natural dynamics of non-hierarchical cognition.

Hierarchical societal demands—such as stillness, waiting one's turn, and planned action—are therefore structurally incompatible with this cognitive organization.

The True Nature of "Hyperactivity"

What is conventionally labeled as *hyperactivity* is more accurately understood as **the bodily expression of continuous network exploration**. Non-hierarchical cognition operates dynamically across networks, and bodily movement synchronizes with this exploration.

Typical manifestations include:

- **Inability to remain still**: continuous bodily exploration of the environment
- **Fidgeting while seated**: micro-level exploratory movements
- **Walking around classrooms or spaces**: spatial network exploration
- **Constant manipulation of objects**: tactile network exploration
- **Continuous speech**: bodily and linguistic processes operate in parallel

The True Nature of "Impulsivity"

What is conventionally labeled as *impulsivity* reflects the **absence of hierarchical inhibition**. In hierarchical cognition, behavior is mediated through sequential stages such as deliberation, evaluation, and execution. In non-hierarchical cognition, perception and action are directly linked within the network, resulting in immediate behavioral output.

Typical manifestations include:

- **Difficulty waiting turns**: waiting presupposes hierarchical temporal structuring
- **Interrupting others**: others' actions are not automatically encoded as higher-priority nodes
- **Risk-taking behaviors**: absence of hierarchical danger-evaluation processes
- **Acting before thinking**: cognition and action occur in parallel
- **Immediate gratification**: lack of hierarchical future-oriented time representation

Expression of Body Dominance

Bodily sensation, movement, and spatial exploration form the core of cognition. Verbal explanation and abstract planning are less effective than physical trial-and-error. Individuals often demonstrate high concentration and competence in physical activities such as sports, dance, or construction, while experiencing significant distress in sedentary, lecture-based environments.

Social Evaluation

Such individuals are frequently labeled as "restless" or "problematic." Natural exploratory bodily activity conflicts with hierarchical social norms emphasizing stillness, order, and planning, leading to moralized misinterpretations such as poor discipline or selfishness.

3.2.2.3 Understanding the Combined Presentation DSM-5 identifies a *combined presentation* in which both inattentive and hyperactive–impulsive features are prominent. In the present theory, this corresponds to a state in which **both linguistic and bodily modalities are non-hierarchically activated**, with dominance fluctuating depending on context rather than remaining fixed.

3.2.2.4 Reinterpreting Executive Dysfunction ADHD has long been conceptualized as a disorder of executive function (Barkley, 1997). In contrast, the present framework argues that ADHD does not reflect a failure of executive function per se, but rather the **presence of a non-hierarchical executive system operating within a hierarchically structured society**.

Non-hierarchical executive functions include:

- **Parallel processing** of multiple information streams
- **Improvisational responsiveness** to unpredictable environments
- **Context-dependent decision making** rather than rule-based judgment
- **Exploratory learning** through trial and error

While these functions are undervalued or penalized in hierarchical societies, they confer significant adaptive advantages in environments requiring exploration, creativity, and rapid adaptation. The core issue in ADHD is therefore not cognitive deficiency, but **structural incompatibility between cognitive organization and social demands**.

3.3 AuDHD (Co-occurring Diagnosis) and the Theory of Hybrid Cognition

3.3.1 Limitations of Conventional Understanding

The co-occurrence of ASD and ADHD, commonly referred to as *AuDHD*, has long posed a theoretical challenge within conventional diagnostic frameworks. Historically, ASD and ADHD were considered mutually exclusive conditions, and even after the DSM-5 permitted co-occurring diagnoses, little theoretical explanation was offered as to **why** these two patterns so frequently coexist.

As a result, AuDHD has often been treated as a mere additive condition—an individual who "has both ASD and ADHD." Such an approach describes symptom overlap but fails to explain the internal cognitive structure that produces the characteristic fatigue, inconsistency, and instability reported by many individuals with this diagnosis.

3.3.2 Reinterpreting AuDHD in This Framework: A Unified Account Based on Moderate Metacognition

In this theoretical framework, the core nature of AuDHD is explained in a unified manner as a condition characterized by a **moderate level of metacognition**. This account refines earlier interpretations that described AuDHD as a combination of a "non-hierarchical cognitive base plus an acquired hierarchical layer," by specifying the metacognitive level at which this hybridization remains incomplete.

3.3.2.1 Characteristics of a "Moderate" Level of Metacognition

At a moderate level of metacognition, individuals **partially understand their own cognitive style but cannot fully regulate or control it**. This incompleteness gives rise to the characteristic presentation of AuDHD: individuals appear to be "functioning" socially, yet experience profound internal exhaustion.

Cognitive characteristics:

- Partial ability to verbalize one's own cognitive style
- Partial understanding of social demands
- Attempts to construct strategies for adaptation (incomplete)
- Translation processes becoming a chronic cognitive burden

3.3.2.2 Non-Hierarchical Cognition × Moderate Metacognition: Incomplete Construction of a Translation Layer

Structure:

Individuals are innately non-hierarchical in their cognition. However, because their metacognition is only moderate, they attempt to construct a translation layer in order to adapt to a hierarchically structured society. As their metacognition has not reached a high level (the "galaxy-type"), this translation layer remains incomplete.

Manifestations of incompleteness:

- **ADHD-like features (non-hierarchical base):**
  - Parallel processing: all information attracts attention equally
  - Associative thinking: ideas emerge in rapid, non-linear leaps
  - Improvisational responsiveness: immediate reactions take precedence over planning
  - Fluid sense of time: immersion in the "now"
- **ASD-like features (attempts at a translation layer):**
  - Preoccupation with rules: explicit learning of hierarchical social rules and attempts to apply them rigidly
  - Systematization attempts: forcing non-hierarchical information into hierarchical structures
  - Social scripts: memorizing and applying "correct" patterns of social interaction
  - Preference for predictability: avoidance of change and strong attachment to routines

Internal conflict:

The natural processing tendencies of non-hierarchical cognition collide with the demands imposed by the translation layer, producing a chronic cognitive burden. Individuals are caught in persistent tension between "what one should do" (the translation layer) and "what one wants to do" (non-hierarchical cognition).

Typical internal experiences include:

- Feeling compelled to follow rules while impulsively deviating from them
- Making plans but quickly becoming distracted by other stimuli
- Knowing the socially "correct" response yet being unable to enact it
- A persistent sense of unease, expressed as "Why am I always so exhausted?"

3.3.2.3 Hierarchical Cognition × Moderate Metacognition: Incomplete Learning of Implicit Knowledge Processing

(Rarer, but present)

Structure:

Individuals are innately hierarchical in their cognition, but because their metacognition is only moderate, they attempt to learn how to process implicit knowledge. As their metacognition does not reach the level associated with typical development, this learning remains incomplete.

Manifestations of incompleteness:

- **ASD-like features (hierarchical base):**
  - Logical thinking: a preference for systematic and structured understanding
  - Explicit rules: attempts to formalize implicit social norms
  - Orientation toward consistency: intolerance of contradiction
- **ADHD-like features (attempts at implicit knowledge processing):**
  - Impulsive social behavior: overreacting while trying to "sense" implicit rules
  - Diffuse attention: attending to everything in search of subtle cues
  - Over-adaptation: exhaustion resulting from excessive efforts to read others' expectations

3.3.2.4 Two Origins of AuDHD This theoretical framework clarifies that AuDHD has two distinct origins:

1. **Non-hierarchical origin** (more common): individuals with non-hierarchical cognition attempt to construct a translation layer
2. **Hierarchical origin** (rarer): individuals with hierarchical cognition attempt to learn implicit knowledge processing

In both cases, AuDHD represents an incomplete attempt at adaptation occurring at a **moderate level of metacognition**.

3.3.2.5 Chronic Accumulation of Translation Cost At a moderate level of metacognition, translation processes become chronic. Because the translation layer cannot be fully automated (as would be possible with high metacognition), continuous conscious effort is required. This chronic translation cost manifests as the characteristic exhaustion associated with AuDHD, including burnout, depressive states, and somatic symptoms.

Accumulation of translation cost:

- Ongoing depletion through everyday social interactions
- Appearing to "function" externally while being near one's internal limit
- Chronic fatigue that does not resolve with rest
- Sudden breakdowns (meltdowns and shutdowns)

3.3.2.6 Why Does Metacognition Remain at a "Moderate" Level? The level of metacognition is a composite outcome of genetic, developmental, and environmental factors.

- **Genetic factors:** individual differences in metacognitive capacity itself
- **Developmental factors:** metacognition tends to improve with development, but may involve critical periods
- **Environmental factors:** supportive environments (which promote self-understanding) versus suppressive environments (which enforce adaptation)

For many individuals with AuDHD, environmental pressures—such as those encountered in school, family, and broader society—may have fixed metacognition at a moderate level. Before reaching full self-understanding (high metacognition), pressures to "adapt" compelled the construction of a translation layer, effectively arresting further metacognitive development.

3.3.3 The Author's Case: A Developmental Path from Non-Hierarchical × Language-Dominant × High Metacognition (Galaxy Type)

The author's long-term self-analysis illustrates a developmental trajectory in which **non-hierarchical × language-dominant cognition** progresses through stages of metacognitive development (low → moderate → high), ultimately reaching the *galaxy type*. This pathway highlights, through contrast with AuDHD, the decisive importance of metacognitive level.

3.3.3.1 Early Childhood (Ages 0–6): The Foundation of Non-Hierarchical × Low Metacognition

The author's innate cognitive structure was non-hierarchical and language-dominant. Information was processed in a parallel and associative manner, with active verbal thinking. However, metacognition was underdeveloped, and there was no awareness of one's own cognitive style.

Manifestations of ADHD-like traits:

- **Hyperactivity:** constant movement and inability to remain still
- **Impulsivity:** immediate action upon arising thoughts
- **Diffuse attention:** interest in everything but difficulty sustaining focus
- **Lack of temporal awareness:** immersion in the "now," with little sense of past or future

This stage corresponds directly to typical ADHD (non-hierarchical × low metacognition).

3.3.3.2 School Age (Ages 7–12): Forced Construction of a Translation Layer and Transition to Moderate Metacognition

Environmental pressure: extreme stress

Severe environmental stress during early childhood—including emotional neglect, linguistic and relational chaos, and economic hardship—created a survival crisis that triggered a rapid transformation of the cognitive structure. In order to survive, it became necessary to learn the rules of a hierarchical society.

Acquisition of a translation layer: an AuDHD-like state

During this period, a hierarchical layer grounded in ethics and logic was acquired secondarily. This process represents a prototypical formation pathway of AuDHD (non-hierarchical × moderate metacognition).

Acquired characteristics:

- **Rule adherence:** explicit learning and rigid application of social rules
- **Attempts at systematization:** forcing non-hierarchical information into hierarchical order
- **Moral rigidity:** strong orientation toward ethical consistency
- **Preference for predictability:** avoidance of change and pursuit of structure

Somatic breakdown: embodiment of translation cost

During the construction of this translation layer, the body broke down. The most symbolic manifestation was fecal incontinence.

Episodes of fecal incontinence (ages 7–13):

- Frequent occurrences at school, on the way to school, and in front of the home entrance
- Both then and now, it remains unclear why endurance was pushed to the limit
- One interpretation: bodily control entered an untranslatable conflict with hierarchical social protocols (procedures required to go to the bathroom during class)
- Cognitive resources were consumed by translation work, intense self-protective instincts, and emotional conflict, causing a breakdown in the **hierarchical execution of bodily control (intentional inhibition and release)**
- The body expressed a refusal to comply with social protocols

These episodes demonstrate that even in non-hierarchical × language-dominant individuals—without a body-dominant modality—bodily control can become a casualty of translation cost. Even for language-dominant types, the body expresses mismatches with social protocols.

Chronic headaches: persistence of translation cost

Chronic headaches, continuing from this period to the present (age 45), can also be interpreted as the embodiment of translation cost. The cognitive burden of dual processing (a non-hierarchical core plus a hierarchical layer) manifests as somatic symptoms.

3.3.3.3 Adolescence (Ages 13–18): Nihilism and Exhaustion at Moderate Metacognition

Dominance of nihilism

A pervasive sense of "nothing matters" prevailed, accompanied by a lack of motivation for growth. This can be understood as exhaustion of the translation layer and a partial reversion to non-hierarchical processing.

Acceleration of rule violation

Unlike the logical critique of rules seen during school age, rule violations in adolescence were emotionally driven acts of rebellion. As the survival motivation sustaining the translation layer weakened, the layer itself became unstable.

Persistence of an AuDHD-like state

Metacognition remained at a moderate level, and the cognitive state continued to be AuDHD-like. The translation layer existed but was incomplete, producing chronic exhaustion.

3.3.3.4 Early Adulthood (Ages 18–36): Social Friction and Depletion "Functioning" AuDHD

While maintaining the translation layer, the author functioned within a hierarchical society (employment, interpersonal relationships). Externally, adaptation appeared successful, but internally there was a sense of emptiness.

Chronic translation cost

Everyday social interactions constantly required translation work. Exhaustion accumulated, yet the reason for this persistent fatigue remained unclear. This state represents a typical adult manifestation of AuDHD at a moderate metacognitive level.

3.3.3.5 Turning Point (Age 36): Transition to High Metacognition and Establishment of the Galaxy Type Acquisition of the self

At age 36, the author acquired a stable sense of self. This signifies a deep understanding of one's cognitive structure and the ability to separate the translation layer from the self, relativizing it as a tool. Metacognition transitioned from moderate to high.

3.3.3.6 Establishment of the Galaxy Type: Non-Hierarchical × Language-Dominant × High Metacognition At this stage, the author became a galaxy type.

Characteristics of the galaxy type:

- Retention of non-hierarchical cognition as the deep structure
- Use of a hierarchical layer at the surface, flexibly deployed according to context
- Translation occurs automatically in the unconscious domain (no conscious effort required)
- Complete understanding and acceptance of one's cognitive style
- Translation cost is greatly reduced, though not entirely eliminated

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3.3.3.7 Differences from AuDHD The decisive difference between moderate metacognition (AuDHD) and high metacognition (galaxy type) lies in control over the translation layer.

Dimension	Moderate Metacognition (AuDHD)	High Metacognition (Galaxy Type)
Translation layer	Incomplete; requires conscious effort	Complete; automated
Self-understanding	Partial	Complete
Internal conflict	Chronic	Resolved
Translation cost	High; accumulative	Low; manageable
Social adaptation	"Functioning" but exhausting	"Functioning" and sustainable

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3.3.3.8 Positioning within the Eight-Quadrant Model When the author's developmental trajectory is mapped onto the eight-quadrant model:

- **Early childhood (0–6):** Non-hierarchical × language-dominant × low metacognition (ADHD, inattentive type)
- **School age (7–12):** Transition toward non-hierarchical × language-dominant × moderate metacognition (AuDHD formation phase)
- **Adolescence–early adulthood (13–36):** Non-hierarchical × language-dominant × moderate metacognition (AuDHD consolidation phase)
- **Post-transition (36–):** Non-hierarchical × language-dominant × high metacognition (galaxy type)

This trajectory demonstrates that the development of metacognition, within the same cognitive structure (non-hierarchical × language-dominant), produces distinct manifestations: ADHD → AuDHD → galaxy type.

3.3.3.9 Reinterpreting Fecal Incontinence: Translational Failure between Bodily Control and Social Protocols

Episodes of fecal incontinence offer critical theoretical insight within this framework.

Somatic symptoms in language-dominant types

The author is language-dominant, not body-dominant. Nevertheless, bodily control (excretion) collapsed. This demonstrates that even when the dominant modality is linguistic, the body expresses translational failure with social protocols.

Mechanism of translational failure

Excretory protocols in hierarchical societies are themselves hierarchical:

1. Recognition of bodily need
2. Assessment of social context (classroom, public space)
3. Execution of permission-seeking procedures (raising one's hand, asking a teacher)
4. Movement to an appropriate location
5. Execution of excretion

For non-hierarchical × moderate-metacognition individuals, this hierarchical sequence requires translation. When the translation layer is incomplete and other cognitive demands (learning, social interaction, emotional stress) are high, bodily control translation collapses.

"Why endure until the limit?"

The answer lies in translational failure. Bodily needs were recognized, but the process of translating them into social protocols failed. As a result, physical limits were reached and control collapsed.

Incontinence as bodily resistance

An alternative interpretation is that the body refused compliance with social protocols. For a non-hierarchical body, hierarchical procedures (permission → movement → execution) are unnatural. The body resisted these demands and responded in a natural mode (need → immediate release).

3.3.3.10 The Present (Age 45): Sustained Galaxy Type and Residual Translation Cost

Automation of translation

At present, the translation layer functions automatically in the unconscious domain. In interactions with hierarchical society, hierarchical language and logic are used without conscious effort.

Residual translation cost

However, translation cost has not disappeared entirely. Prolonged social interaction, complex hierarchical tasks, and unpredictable social situations still produce exhaustion. Chronic headaches persist.

Limits of the galaxy type

Even with high metacognition, translation cost cannot be reduced to zero. Translating non-hierarchical cognition into hierarchical formats is inherently burdensome. The galaxy type can manage and sustain this burden, but cannot eliminate it.

Implications for Structural Differentiation

This limitation underscores the necessity of **Structural Differentiation**. To fundamentally reduce translation cost, non-hierarchical individuals must cease living within hierarchical societies and instead inhabit environments optimized for non-hierarchical cognition. Even for galaxy types, complete liberation from translation cost is only possible through structural differentiation.

3.3.3.11 Theoretical Significance of the Author's Case

The author's case demonstrates:

- **Stages of metacognitive development:** manifestations vary by metacognitive level even within the same cognitive structure
- **Mechanism of AuDHD formation:** environmental pressure fixes metacognition at a moderate level
- **Possibility of transition to the galaxy type:** movement to high metacognition is possible under appropriate conditions
- **Somatization of translation cost:** bodily symptoms (incontinence, headaches) are traces of structural violence
- **Necessity of Structural Differentiation:** even high metacognition does not fully eliminate translation cost

3.3.4 Implications for Other Diagnoses

The present theoretical framework also suggests new possibilities for understanding diagnoses beyond ASD and ADHD.

Borderline Personality Disorder (BPD) may be conceptualized as the interaction of **non-hierarchical cognition and relational disruption (trauma)**. For individuals with non-hierarchical cognition, interpersonal relationships constitute the foundational network through which the self is organized. When attachment instability or traumatic experiences lead to the rupture of relational networks, the cognitive and experiential foundation of the self collapses. Consequently, fear of abandonment may be experienced not merely as fear of rejection, but as the threat of self-annihilation—a form of existential fear that is particularly characteristic of non-hierarchical cognition.

Obsessive–Compulsive Disorder (OCD) may be understood as an excessive refinement of **hierarchical cognition**. As a defensive response to environmental uncertainty, individuals attempt to preserve cognitive structure through ritualization and repetitive checking. Intrusive thoughts represent information that cannot be integrated into an existing hierarchical structure; compulsive behaviors emerge as attempts to eliminate, neutralize, or contain these structurally incompatible elements.

These interpretations are provisional and intended as theoretical suggestions rather than definitive accounts. Further systematic and empirical investigation will be required to evaluate the validity and scope of these extensions.

4 Translation Cost and Structural Asymmetry

In the previous chapter, this paper examined how differences in cognitive types are manifested within existing diagnostic categories. The present chapter analyzes how these differences give rise to specific problems through their interaction with social structures. The core concept introduced here is **translation cost**.

4.1 The Concept of Translation Cost

4.1.1 Definition

Translation cost refers to the cognitive, temporal, and emotional burdens that arise when individuals with different cognitive structures communicate with one another or attempt to adapt to a shared environment. Just as communication between speakers of different languages requires one party to translate into the other's language, interaction between individuals with different cognitive structures requires the translation of cognitive styles.

This translation is not a mere conversion of linguistic expressions. Rather, it entails transforming the underlying **principle of information organization** itself. For non-hierarchical individuals to function within a hierarchically structured society, they must translate their network-based cognition into hierarchical forms. Conversely, for hierarchical individuals to function in non-hierarchical environments, they must translate their hierarchical cognition into network-based forms.

4.1.2 Directionality of Translation

A critical feature of translation cost is its **directional asymmetry**. Translation from hierarchical to non-hierarchical forms is relatively easy. It requires only the loosening or flattening of hierarchical structures. In general, dismantling a structure is easier than constructing one.

In contrast, translation from non-hierarchical to hierarchical forms is extremely demanding. Information that exists as a network must be reorganized into a hierarchical structure. This process is equivalent to reconstructing innumerable nodes and links in a network into a tree structure defined by logical inclusion and sequential order. Such reconstruction imposes a chronic cognitive overload on non-hierarchical individuals.

4.1.3 Components of Translation Cost

Translation cost consists of multiple interrelated components.

Cognitive burden refers to the occupation of working memory required for translation. When non-hierarchical individuals function within a hierarchical society, they must continuously perform the parallel task of translating their cognition into hierarchical forms. This translation process consumes substantial working-memory resources, thereby reducing the cognitive capacity available for other activities such as learning, creativity, and problem solving.

Temporal cost refers to delays in processing time caused by translation. Tasks that hierarchical individuals can understand and execute immediately must be processed via an additional translation step by non-hierarchical individuals. As a result, processing speed decreases. This is often misinterpreted as "inattention" or "slowness," whereas in reality it reflects the presence of an additional cognitive process.

Emotional cost refers to the stress, fatigue, and self-negation that accompany translation. Suppressing one's natural cognitive style and being forced to operate in an unnatural mode generates psychological distress. When translation is incomplete, individuals may further experience self-blame ("Why can't I do this?") and criticism from others.

Although each of these costs may appear minor in the short term, their chronic accumulation can have severe consequences. Over time, accumulated translation cost manifests as physical symptoms, psychological exhaustion, and social maladaptation.

4.2 Structural Asymmetry of Hierarchical Society

Modern society is optimized for hierarchical cognition at every structural level. As a result, individuals with non-hierarchical cognition are compelled to engage in continuous translation across all layers of society.

4.2.1 Language and Information Structure

Language itself is a product of hierarchical cognition. The subject–predicate structure presupposes a hierarchical relationship in which the subject occupies a higher-level position and the predicate a subordinate one. Tense organizes time into a linear hierarchy of past, present, and future. Likewise, the relationship between abstract and concrete concepts is represented hierarchically, with abstract concepts positioned above their concrete instances.

For individuals with non-hierarchical cognition, articulating lived experience through language is structurally difficult. In non-hierarchical cognition, information exists simultaneously and in parallel, whereas language unfolds in a linear and sequential manner. The hierarchical explanatory style commonly demanded in social contexts—"first the conclusion, then the reasons"—is experienced as unnatural by non-hierarchical individuals, whose cognition does not privilege linear ordering or top-down structuring.

4.2.2 Institutions and Organizations

Social institutions are themselves hierarchical structures. Corporations, schools, and bureaucratic systems define authority, roles, and affiliation according to one's position within a hierarchy. For individuals with non-hierarchical cognition, these hierarchies are often experienced as artificial. Because they perceive relationships as networks, parallel relations such as "collaborators" or "project members" feel more natural than hierarchical relations such as "superiors" and "subordinates."

However, within hierarchical organizations, behavior that ignores or bypasses formal hierarchies is often sanctioned as "disrespectful" or as a "violation of rules." In this way, non-hierarchical modes of interaction are systematically penalized, reinforcing the asymmetry between cognitive types.

4.2.3 Economic Systems

Capitalist economic systems are grounded in long-term planning, accumulation, and predictability—features that align closely with the characteristics of hierarchical cognition. Organizational structures (pyramidal hierarchies), career paths (stepwise promotion), and financial systems (savings and compound interest) are all based on principles of systematic organization and hierarchical accumulation.

Individuals with non-hierarchical cognition often prioritize immediacy over accumulation and the present over the future, and therefore tend to experience difficulty with capitalist expectations of planning and saving. This tendency does not reflect irrationality but rather a different form of rationality: maximizing value by releasing available resources—such as creativity and responsiveness—in the present environment, and optimizing utility within ongoing flows rather than through long-term hierarchical accumulation.

4.2.4 Education

Formal education is a paradigmatic example of a hierarchical system. Curricula are designed to construct knowledge step by step, and evaluation is conducted through hierarchical metrics such as scores, grades, and rankings. For non-hierarchical children, this sequential and systematic mode of learning is often experienced as painful.

Non-hierarchical learners tend to prefer exploratory and discontinuous learning driven by interest, moving jumpwise rather than incrementally. Hierarchical education systems, however, typically reject this learning style by insisting that "application is impossible without mastery of the basics," thereby denying the legitimacy of non-hierarchical modes of learning.

4.2.5 Medicine

Medical systems are also hierarchically structured. Diagnostic frameworks such as DSM-5 and ICD-11 classify conditions through hierarchical taxonomies, and treatment proceeds through a hierarchical process of diagnosis, treatment planning, and implementation.

More importantly, medicine positions "health" as a higher-order state and "disorder" as a lower-order deviation, implicitly defining health as the capacity to function within a hierarchical society. In this sense, medical systems operate as institutional mechanisms that attempt to convert non-hierarchical cognition into hierarchical forms. Rather than merely treating suffering, medicine often functions to enforce adaptation to hierarchical social norms.

4.3 Components of Translation Cost

4.3.1 Asymmetry of Translational Burden

The hierarchical structure of contemporary society, as described in the previous section, generates an asymmetry in translation cost. This asymmetry functions as a form of **structural violence**.

In modern society, translation cost is borne almost exclusively by non-hierarchical individuals. Because all layers of society are optimized for hierarchical cognition, individuals with hierarchical cognition are not required to translate. In contrast, non-hierarchical individuals are forced to translate their cognition in virtually every social context. This unilateral imposition of burden is not a matter of individual ability, competence, or effort.

Rather, it constitutes a form of structural injustice: **a violation of cognitive rights, in which one cognitive type is compelled to adopt the language and structure of another in order to survive**. Much of this burden cannot be eliminated through individual effort, self-improvement, or medical treatment. The reduction of translation cost is achievable only through the redesign of social structures, not through the modification of individuals.

4.3.2 Social Pressure ("Noise")

Translation cost is not limited to the cognitive effort required for translation. Hierarchical society also imposes the internalization of its value system upon non-hierarchical individuals. This is experienced as the constant reception and attempted application of **values that contradict one's own cognitive structure**. In this paper, this phenomenon is referred to metaphorically as *"noise."*

Be planful. Be responsible. Be punctual. Prioritize. Regulate your emotions. All of these norms presuppose hierarchical cognition. Non-hierarchical individuals receive these messages continuously and attempt to apply them to themselves. However, because such norms are structurally incompatible with their cognition, this process repeatedly results in the experience of "failure," producing chronic self-negation.

More critically, this social pressure occupies working memory. Non-hierarchical individuals would otherwise allocate cognitive resources to creative activity, learning, or problem-solving. Instead, these resources are consumed by the ongoing need to monitor, interpret, and respond to socially imposed norms.

In the author's case, a fundamental indifference toward social values prevents the reception of this noise. Social norms do not constitute "salient nodes" within the author's galaxy-like cognitive structure and therefore are not captured by its gravitational field. However, this is an unusual trait. Most non-hierarchical individuals remain continuously exposed to social noise.

4.3.3 "Adaptation" as Assimilation

What is labeled as "support" within medicine, education, and welfare is often, in practice, a form of assimilation pressure toward hierarchical cognition. The term *adaptation* appears neutral, but its substantive meaning is alignment with hierarchical society.

Pharmacological and behavioral interventions alike aim to increase conformity to hierarchical norms. These forms of support implicitly convey the message: *"Your cognitive style is incorrect; you must change to become hierarchical."* However, cognitive structure is fundamental and cannot be easily altered.

Genuine support does not consist in assimilation into hierarchical cognition, but in the provision of environments in which non-hierarchical individuals can live while preserving their own cognitive structure.

4.4 Embodiment of Translation Cost

When translation cost accumulates chronically, it manifests as physical symptoms, psychological distress, and social consequences. These manifestations are not "symptoms of disability," but traces of structural violence.

4.4.1 Physical Symptoms

Translation cost frequently manifests as physical symptoms. Chronic headaches, incontinence, autonomic dysregulation, and sleep disorders are commonly reported. These symptoms indicate that cognitive overload can extend into bodily regulation.

In one case, chronic headaches persisting from childhood into the present (age 45) were reported. These headaches can be interpreted as the bodily embodiment of sustained cognitive overload caused by dual processing: performing translation work while operating within a hierarchical society.

In the same case, frequent incontinence (both urinary and fecal) occurred throughout childhood and continued until age thirteen. These episodes occurred in various locations (school, on the way home, at the entrance of the home), and the individual reported an inability to understand why they endured the urge until the limit was exceeded. One interpretation is that cognitive resources were consumed by translation work and emotional burden (including a harsh family environment), leaving insufficient resources for bodily regulation. The body persistently refused adaptation to hierarchical society.

4.4.2 Psychological Symptoms and Social Consequences

The psychological consequences of translation cost manifest as nihilism, burnout, depression, and anxiety. Continuous suppression of one's natural cognitive style and the imposition of an unnatural one entail severe psychological distress. Nihilism during adolescence can be understood as a loss of interest in the "game" of hierarchical society, reflecting exhaustion of the translation layer.

Withdrawal from translation is socially observed as "maladaptation." Social withdrawal, job loss, and poverty are regarded as failures within hierarchical society, but in fact represent bodily and psychological responses to translational impossibility. These physical symptoms, psychological states, and social outcomes are not indicators of individual deficiency or maladaptation; they are the accumulated consequences of translation cost imposed by hierarchical society on non-hierarchical cognition. The problem resides not in individuals, but in social structure.

5 The Proposal for Cognitive Segregation

In the preceding chapters, this paper has examined differences in cognitive types, the ways in which these differences are expressed through diagnostic categories, and the translation costs imposed on non-hierarchical cognition by hierarchical society. In this chapter, **cognitive segregation** is proposed as a response to this structural problem.

5.1 Problems with the Coexistence Model

The dominant ideal in contemporary society is *inclusion*. The Convention on the Rights of Persons with Disabilities, the neurodiversity movement, and diversity-oriented management all promote the vision that diverse individuals should coexist within the same social spaces. This ideal emerged as a progressive reaction against historical practices of exclusion and segregation. However, in the context of cognitive-type differences, the coexistence model contains fundamental flaws.

5.1.1 Inclusion as Assimilation Pressure

The term *inclusion* suggests equal coexistence, but in practice it often means assimilation into hierarchical society. Inclusion typically refers to incorporating diverse individuals into existing social structures while leaving those structures fundamentally unchanged. Because these structures are optimized for hierarchical cognition, non-hierarchical individuals are continuously forced to translate their cognition in order to function within them.

Reasonable accommodations likewise represent only limited adjustments that presuppose existing structures. Extended examination time, quieter environments, or visual supports may be helpful, but they do not alter the foundational architecture of hierarchical society—such as curricula, evaluation systems, or organizational hierarchies. As a result, non-hierarchical individuals continue to pay translation costs. Reasonable accommodation thus creates a persistent trade-off between the individual and society rather than resolving the underlying problem.

The rhetoric of "respecting diversity" is often hollow. Hierarchical society may claim to respect non-hierarchical individuals while simultaneously demanding that they become more hierarchical. Adaptation training, social skills training, and pharmacological interventions all function as attempts to convert non-hierarchical cognition into hierarchical forms.

5.1.2 Structural Incompatibility of Cognitive Architectures

The core problem of the coexistence model is its failure to recognize the structural incompatibility between hierarchical and non-hierarchical cognition. This is not a matter of stylistic preference or individual inclination; it is a difference at the most fundamental level of information organization.

When cognition is hierarchized, non-hierarchical individuals cannot function optimally. Organizing information hierarchically, prioritizing tasks, and processing information in stepwise sequences are inherently unnatural and cognitively costly for non-hierarchical cognition. Conversely, when cognition is flattened, hierarchical individuals struggle. Information that exists in parallel without stable hierarchy or fixed structure produces confusion and anxiety for hierarchical cognition.

Attempts to "respect both sides" often result in the partial suppression of both. Requiring individuals to switch between hierarchical and non-hierarchical modes depending on context imposes translation costs on both cognitive types. The outcome is an environment in which neither type can function optimally.

5.1.3 The Persistence of Translation Cost

Within the coexistence model, translation cost is never eliminated. As long as hierarchical social structures remain intact, non-hierarchical individuals must continue translating their cognition in every domain of life. Even in "understanding environments" or "accommodating workplaces," the foundational layers of society—language, institutions, and evaluation systems—remain hierarchical.

Moreover, the coexistence model renders the asymmetry of translation cost invisible. The absence of translation effort by hierarchical individuals is not problematized, while the translation performed by non-hierarchical individuals is praised as "adaptation." As long as this asymmetry persists, structural injustice is reproduced.

Coexistence is rhetorically appealing, but in practice it preserves hierarchical privilege. While claiming to include diversity, it maintains hierarchical structures and imposes unilateral burdens on non-hierarchical individuals. Genuine respect for diversity requires enabling different cognitive types to live in environments optimized for their own cognitive structures. **That is the rationale for cognitive segregation.**

5.2 The Proposal of the Cognitive Segregation Model

5.2.1 Fundamental Principles

Cognitive segregation refers to a social arrangement in which each cognitive type lives in environments optimized for its own cognitive structure. This is a form of separation, but **not segregation in the historical sense of oppression**. It is neither coercive nor exclusionary. Rather, it is based on voluntary choice. Each type selects the environment that is most natural to it—one that minimizes translation cost—and lives, creates, and contributes within that environment.

The core claim of cognitive segregation is that **translation should be minimized**. The coexistence model assumes that individuals must continuously translate their cognition to fit a shared social space. This expectation is unsustainable and undermines human dignity. Cognitive structure constitutes a fundamental dimension of personhood. To constantly suppress it and enforce an unnatural mode of functioning is a form of violence. Cognitive segregation represents liberation from this violence.

Through segregation, each cognitive type can fully realize its capacities. Hierarchical types excel at systematic knowledge construction, institutional design, and long-term planning. Non-hierarchical types excel at creativity, adaptability, and immediate responsiveness to situations. However, these capacities emerge only within appropriate environments. Placing hierarchical individuals in non-hierarchical environments produces confusion; placing non-hierarchical individuals in hierarchical environments produces exhaustion. Cognitive segregation enables each type to live within conditions suited to its cognitive nature.

5.2.2 Spatial Segregation

Spatial segregation refers to the construction of physically distinct environments. Hierarchical cities and non-hierarchical communities coexist in parallel.

Hierarchical cities emphasize planned urban design, hierarchical organizations, and systematic public services. Residents value predictability, order, and structure. Buildings are arranged systematically, transportation follows schedules, and administration operates bureaucratically.

Non-hierarchical communities emphasize fluid use of space, project-based collaboration, and improvisational mutual support. Residents value flexibility, exploration, and relationality. Spaces are reconfigured as needed, activities change in response to context, and decision-making occurs through networks rather than chains of command.

This spatial segregation does not imply complete disconnection. Movement, visits, and interaction between environments remain possible. However, the foundation of everyday life is situated within environments aligned with one's own cognitive structure.

5.2.3 Institutional Segregation

Institutional segregation involves designing social institutions—such as education, labor, and economic systems—according to cognitive types.

In education, curriculum-based education for hierarchical types coexists with exploratory education for non-hierarchical types. The former delivers knowledge sequentially and systematically and evaluates performance through examinations. The latter allows for non-linear, interest-driven learning and evaluates outcomes through projects and portfolios. Children can select educational pathways aligned with their cognitive structure.

In labor systems, organization-based work for hierarchical types coexists with project-based work for non-hierarchical types. The former assumes corporate structures, formal positions, and long-term employment. The latter assumes freelance work, collaborative projects, and short-term contracts. Individuals choose work styles consistent with their cognitive type.

In economic systems, monetary economies for hierarchical types coexist with gift-based or relational economies for non-hierarchical types. The former emphasizes accumulation, investment, and planned consumption. The latter emphasizes immediacy, sharing, and relationship-based exchange. These systems may interact at interfaces, but function according to different principles in everyday life.

5.2.4 Informational Segregation

Informational segregation refers to designing communication styles and knowledge-management systems according to cognitive types.

Hierarchical communication relies on documents, reports, and layered approval processes. Information is systematically organized and transmitted from higher to lower positions. Non-hierarchical communication relies on dialogue, improvisation, and network-based diffusion. Information circulates within relationships and is restructured as needed.

Knowledge management follows similar patterns. Hierarchical types use libraries, databases, and classification systems. Non-hierarchical types use networks, associative connections, and dynamic linking. Each type is thus able to engage with information in ways that feel cognitively natural.

Cognitive segregation operates across these three dimensions—spatial, institutional, and informational. Crucially, this is not exclusion but optimization. It involves constructing environments in which each cognitive type can fully express its strengths. In this sense, cognitive segregation constitutes a genuine respect for diversity.

5.3 Designing Interfaces Between Segregated Systems

Cognitive segregation does not imply complete disconnection. Interfaces between hierarchical societies and non-hierarchical societies are necessary. This section examines how such interfaces should be designed.

5.3.1 Clear Boundaries

For segregation to function effectively, boundaries must be explicit. Where does hierarchical space begin, and where does non-hierarchical space end? These boundaries must be clarified not only physically (e.g., urban borders, building entrances) but also institutionally (e.g., which educational system one belongs to, which labor model one adopts) and informationally (e.g., which communication style is used).

These boundaries are not intended for exclusion but for protection. When a non-hierarchical individual enters a hierarchical space, they are aware in advance that translation will be required. Conversely, when a hierarchical individual enters a non-hierarchical space, they understand that hierarchical expectations—such as strict punctuality, long-term planning, and explicit rule enforcement—may not apply. Boundaries thus provide an opportunity for cognitive preparation, helping to avoid unexpected translation costs.

Boundaries are not fixed. Individuals may cross them depending on life stage, circumstances, or personal choice. However, such crossings are voluntary and made with an understanding of the translation costs involved.

5.3.2 The Role of Translators

At these boundaries, translators are required. A translator is someone who understands both cognitive types and can mediate between them. In many cases, hybrid types naturally assume this role.

Hybrid types possess both a non-hierarchical cognitive core and a hierarchical cognitive layer, enabling them to navigate and interpret both worlds. They can explain the logic of hierarchical societies to non-hierarchical individuals and translate non-hierarchical experiences into forms intelligible to hierarchical systems. In contexts where the two societies come into contact—such as commercial transactions, legal procedures, or joint projects—translators are indispensable.

However, the role of the translator is not to translate continuously. Translation occurs only at necessary points of contact. In daily life, each cognitive type primarily inhabits its own optimized environment, dramatically reducing the frequency of translation. Translators thus function as specialized professionals within a segregated society and should be appropriately recognized and compensated.

5.3.3 Selective Interaction

Segregation does not prohibit interaction. On the contrary, it enables interaction to become voluntary and meaningful.

When hierarchical individuals visit non-hierarchical communities, they can experience creativity, improvisation, and relational richness. When non-hierarchical individuals visit hierarchical cities, they can learn from systematic knowledge, predictability, and structural stability. Crucially, these are visits rather than permanent residence—short-term engagements that foster mutual understanding without imposing chronic translation costs.

Selective interaction is also important in education. Non-hierarchical children may temporarily experience hierarchical education to gain insight into hierarchical logic, and vice versa. However, everyday education remains grounded in environments aligned with each child's cognitive type.

Cultural, academic, and artistic exchanges likewise flourish in a segregated society. As each society develops its own distinctive culture, interactions become more stimulating. Hierarchical societies generate systematic scholarship and precise artistic forms, while non-hierarchical societies produce improvisational expression and exploratory creativity. Their interaction enhances humanity's overall cultural richness.

Segregation does not entail isolation. Through clear boundaries, translators, and selective interaction, segregated societies coexist. Whereas coexistence models demand continuous, compulsory translation, segregation enables voluntary and sustainable exchange.

5.4 Ethical Considerations and Feasibility

The proposal of cognitive segregation requires careful attention to historical context and a serious examination of feasibility. This section addresses these concerns.

5.4.1 Ethical Considerations

The term "separation" inevitably evokes negative historical associations. Racial segregation, segregated education, and institutional confinement have all been used as tools by majorities to exclude and oppress minorities. Given this history, skepticism toward segregation is both understandable and necessary.

However, the form of cognitive segregation proposed in this paper is fundamentally different from historical segregation.

First, segregation in this framework is **voluntary rather than coercive**. Individuals are free to choose which social environment they belong to, based on their cognitive type and life needs. Freedom of movement, boundary crossing, and interaction is guaranteed.

Second, cognitive segregation does **not presuppose a hierarchy of superiority or inferiority**. Historical segregation was grounded in the assumption that the majority was "superior," relegating minorities to "inferior" spaces. By contrast, this model assumes that cognitive types are **equivalent in value**, each possessing distinct strengths. There is no ranking in which hierarchical cognition is superior and non-hierarchical cognition inferior.

Third, cognitive segregation presupposes **fair resource allocation**. In historical segregation, minority spaces were systematically deprived of resources. In a segregated cognitive society, each social environment receives resources appropriate to its structure and needs: systematic infrastructure for hierarchical societies, and flexible, fluid resources for non-hierarchical societies.

Cognitive segregation is therefore not a tool of oppression but a **means of liberation**. Forcing non-hierarchical individuals into hierarchical social structures under the banner of "inclusion," while imposing continuous translation, constitutes structural violence.

This demand is not merely about efficiency. It is about the **restoration of self-determination over one's mode of existence**. As argued in Chapter 4, translation cost represents a **violation of cognitive rights**. The demand to escape this violation—the plea of "let us be segregated"—is an **ethical and existential claim for survival**. Accordingly, this paper argues that cognitive segregation represents one of the most ethically grounded forms of social design aimed at liberation from structural violence.

5.4.2 Feasibility

While cognitive segregation may be compelling in principle, the question remains: is it feasible? The answer proposed here is **gradual implementation**.

The first stage involves **small-scale experiments**. In specific regions, schools, or workplaces, the principles of cognitive segregation can be introduced experimentally. Examples include schools practicing exploratory education, organizations centered on project-based labor, and communities prioritizing non-hierarchical communication. Through these experiments, outcomes such as reduced translation cost, enhanced capability expression, and improved well-being can be empirically observed.

The second stage involves **connecting with existing movements**. Homeschooling, alternative education, freelance work, coworking spaces, rural migration, and intentional communities already function as partial withdrawals from hierarchical society. Reinterpreting these phenomena as early forms of cognitive segregation—and supporting their expansion—can help establish the foundation of a segregated cognitive society.

The third stage is **institutional recognition**. If cognitive segregation demonstrates sustained benefits, governmental and municipal support may follow. This could include public recognition of multiple educational tracks, legal protection for diverse employment arrangements, and the circulation of multiple economic systems or currencies.

The primary obstacle to implementation is the **vested interests of hierarchical society**. Hierarchical systems benefit from incorporating non-hierarchical individuals through "inclusion," extracting their labor, creativity, and consumption. Cognitive segregation threatens this extraction by enabling independence, and resistance should therefore be expected.

However, in the long term, segregation benefits both sides. Hierarchical individuals are relieved of the costs associated with managing, educating, or treating non-hierarchical individuals. When each cognitive type operates in an environment optimized for its structure, overall human productivity and creativity increase. Cognitive segregation is not a zero-sum solution, but a **positive-sum outcome**.

5.4.3 Responses to Anticipated Criticisms

Several objections can be anticipated.

To the claim that "segregation will deepen social division," this paper responds that division already exists. Hierarchical and non-hierarchical individuals are already separated at the level of cognitive structure. Forced coexistence under the rhetoric of inclusion merely conceals this division while imposing translation costs on non-hierarchical individuals. Segregation makes the division explicit and enables both types to live with dignity.

To the claim that "segregation is unrealistic," this paper points to the path of gradual realization. Small-scale experiments already exist, connections with current movements are feasible, and institutional recognition can follow. Complete segregation may be a distant goal, but partial segregation is already occurring.

To the claim that "segregation harms children," this paper responds that the current single-system model already inflicts serious harm on non-hierarchical children, including enuresis, chronic headaches, school refusal, and pervasive self-negation. Allowing children to receive education aligned with their cognitive type enhances, rather than diminishes, child welfare.

Cognitive segregation is a challenging proposal. Yet if the goal is to end translation cost as a form of structural violence, a fundamental redesign of social structure is unavoidable. Cognitive segregation represents one viable path toward that end.

6 Discussion

This chapter acknowledges the limitations of the present theory, clarifies its falsifiability, outlines directions for future research, and discusses its broader social implications.

6.1 Limitations of the Theory

The present theory has important limitations. Explicitly articulating these limitations is essential for the intellectual integrity of the theory and for its future development.

6.1.1 Lack of Empirical Data

The most significant limitation of this theory is the lack of empirical data. The present paper is based primarily on the author's long-term self-analysis. It constitutes an $n = 1$ first-person study, and its generalizability is therefore limited. There is no guarantee that the author's experience represents a typical case of the non-hierarchical $\times$ language-dominant $\times$ high-metacognition (galaxy-type) configuration. It is entirely possible that other non-hierarchical individuals, or individuals located in other quadrants, possess different cognitive structures or developmental trajectories.

Moreover, this paper does not employ systematic dialogical or interview data. Experiences of other individuals have not been collected or analyzed in a structured manner. As a result, the paper cannot provide empirical evidence regarding the diversity of cognitive structures across the eight quadrants, differences in the manifestation of dominant modality, developmental pathways associated with different levels of metacognition, individual variability in translation cost, or the effects of segregated environments.

Future research must address these gaps through the following empirical approaches:

Collection of Multiple Case Studies (Qualitative Research)

- Detailed descriptions of multiple cases from each quadrant (particularly the eight primary positions)
- Systematic comparisons of how dominant modality (language-dominant vs. body-dominant) manifests in lived experience
- Differences in developmental trajectories and adaptive strategies associated with high, moderate, and low levels of metacognition
- Diversity in experiences of translation cost, including bodily symptoms, psychological symptoms, and social consequences
- Collection of experiential data from environments optimized for non-hierarchical cognition (segregated environments)

Questionnaire-Based Studies (Quantitative Research)

- Development and validation of scales to measure cognitive structure (hierarchical vs. non-hierarchical)
- Development of scales to assess dominant modality (language-dominant vs. body-dominant)
- Development and validation of instruments to measure levels of metacognition
- Large-scale surveys to estimate the distribution of individuals across the eight quadrants
- Empirical testing of the correspondence between existing diagnoses (ASD, ADHD, AuDHD) and the eight-quadrant model
- Examination of relationships among translation cost, well-being, creativity, and social adaptation

Experimental Validation (Cognitive Tasks and Neuroimaging)

- Development of cognitive tasks that distinguish hierarchical processing from network-based processing
- Tasks designed to assess language dominance versus bodily dominance
- Establishment of objective methods for measuring levels of metacognition
- Experimental paradigms for measuring translation cost (e.g., dual-task paradigms)

Of particular importance are the following unresolved issues:

1. **Measurement and Developmental Mechanisms of Metacognition**
    - Objective criteria for distinguishing high, moderate, and low levels of metacognition
    - Factors that lead to the stabilization of "moderate" metacognition (environmental, genetic, developmental)
    - Conditions that facilitate transitions from moderate to high metacognition
    - The existence of critical periods and limits of plasticity
2. **Neural Basis and Plasticity of Dominant Modality**
    - Differences in brain structure and functional connectivity between language-dominant and body-dominant individuals
    - Developmental origins of modality dominance (genetic vs. environmental contributions)
    - The degree of plasticity in modality dominance (possibility of switching between language and bodily dominance)
    - Correspondence between modality dominance and diagnostic categories (e.g., language-dominant ASD vs. body-dominant ASD)
3. **Independence and Interaction of the Three-Dimensional Model**
    - Whether cognitive structure, dominant modality, and metacognition constitute independent dimensions
    - Interactions among dimensions (e.g., whether hierarchical cognition is associated with higher metacognition)
    - Whether the eight quadrants represent discrete categories or a continuous spectrum
    - Intra-individual variability across contexts and developmental stages

6.1.2 Unresolved Neuroscientific Foundations

Although the present theory posits fundamental differences in cognitive structure, its neuroscientific foundations remain unresolved. It is currently unclear how hierarchical versus non-hierarchical cognition corresponds to differences in brain structure, neurotransmitter systems, or large-scale network properties.

Connections between this theory and existing neuroscientific research on ASD and ADHD have not yet been established. Future work must examine how the proposed cognitive types relate to existing neuroimaging findings. Empirical studies using functional MRI, structural MRI, and diffusion tensor imaging (DTI) will be necessary to elucidate the neural substrates underlying hierarchical and non-hierarchical cognition.

6.1.3 Lack of Established Measurement Methods

To empirically test the present theory, reliable methods for measuring cognitive type are required. At present, however, no standardized questionnaires or cognitive tasks exist to distinguish hierarchical from non-hierarchical cognition.

Future methodological development should proceed along several lines. First, self-report questionnaires should be developed to assess information organization principles, processing styles, and everyday cognitive activities, followed by rigorous evaluation of their reliability and validity. Second, experimental cognitive tasks should be designed to measure whether hierarchical or network-based processing is dominant. Third, behavioral observation methods should be established to infer cognitive type from patterns of behavior in naturalistic or experimental settings.

Only after such measurement tools are developed will rigorous empirical testing of the present theory become possible.

6.2 Falsifiability

For a theory to be scientific, it must be falsifiable. This section specifies the kinds of empirical evidence that would warrant the rejection of the present theory.

6.2.1 Conditions for Rejection

The present theory would be rejected if the following types of evidence were obtained.

First, if the distinction between hierarchical and non-hierarchical cognition were entirely undetectable at the neuroscientific level. If no systematic differences corresponding to hierarchical versus non-hierarchical cognition were found in brain structure, functional connectivity, or neurotransmitter systems, the proposed neuroscientific basis of the theory would be undermined.

Second, if large-scale empirical studies revealed no systematic association between cognitive types and diagnostic categories. If no statistically significant relationship were found between ASD and hierarchical cognition, or between ADHD and non-hierarchical cognition, the core claim of the theory would be falsified.

Third, if the existence of translation cost could not be empirically demonstrated. If individuals with non-hierarchical cognition living in hierarchically structured societies did not exhibit increased cognitive load, stress, or physical symptoms, the concept of translation cost would lack empirical support.

Fourth, if the proposed benefits of cognitive segregation were not observed. If individuals with non-hierarchical cognition showed no improvement in well-being, creativity, or productivity when living in environments optimized for non-hierarchical cognition, the proposal of segregation would not be justified.

6.2.2 Competing Hypotheses

Several competing hypotheses can be formulated in opposition to the present theory.

Alternative Hypothesis 1: Mere Individual Differences The dichotomy between hierarchical and non-hierarchical cognition is artificial, and cognitive variation is better understood as continuous individual differences. Cognitive styles are multidimensional and cannot be adequately captured by a single axis such as hierarchy.

Alternative Hypothesis 2: Existing Models Are Sufficient The phenomena addressed by the present theory can be sufficiently explained by existing diagnostic frameworks (e.g., ASD and ADHD) or by established models such as systemizing–empathizing theory. A new theoretical framework is therefore unnecessary.

Alternative Hypothesis 3: Primacy of Environmental Factors Differences in cognitive structure are not innate but are primarily shaped by environmental factors such as education, culture, and socioeconomic status. From this perspective, the concept of stable cognitive types is inappropriate.

Alternative Hypothesis 4: Translation Cost Is Adaptable The difficulties experienced by individuals with non-hierarchical cognition in hierarchical societies can be overcome through appropriate interventions, such as pharmacological treatment, cognitive-behavioral therapy, or environmental accommodations. Radical social restructuring in the form of cognitive segregation is therefore unnecessary.

By empirically testing these competing hypotheses and comparing them with the present theory, the validity of the framework can be evaluated. Science advances through competition among theories, and the present theory should be subjected to such scrutiny.

6.3 Future Directions for Research

To further develop the present theory, research is required in multiple complementary directions.

6.3.1 Empirical Research

The first direction is empirical research, aimed at testing the core claims of the theory using experiential data.

Qualitative Research: Detailed Descriptions of the Eight Quadrants

Through interviews with multiple participants (ASD, ADHD, AuDHD, and non-diagnosed individuals), future studies should explore the characteristics of cognitive structures, experiences of translation cost, and orientations toward cognitive segregation at each position within the eight-quadrant model.

Key focal themes include:

- **Expression of dominant modality**: How the same cognitive structure (e.g., hierarchical cognition) manifests differently in language-dominant versus body-dominant individuals
- **Qualitative differences in metacognitive level**: How "high," "moderate," and "low" metacognition differ experientially
- **Diversity of translation costs**: Individual differences in cognitive burden, bodily symptoms, and emotional exhaustion
- **Two origins of AuDHD**: Differences between non-hierarchical-origin and hierarchical-origin hybrid experiences
- **Formation of the galaxy type**: Conditions enabling the transition from moderate to high metacognition

Recommended interview methods include:

- Semi-structured interviews (2–3 hours)
- Encouraging detailed verbalization of cognitive processes
- Collection of concrete episodes (school, workplace, interpersonal contexts)
- Documentation of bodily manifestations of translation cost (e.g., headaches, incontinence, fatigue)

Quantitative Research: Measurement and Validation of the Eight-Quadrant Model

Large-scale surveys should be conducted using newly developed instruments designed to measure cognitive types.

Development of Measurement Scales

1. **Cognitive Structure Scale (Hierarchical vs. Non-Hierarchical)**
    - Mode of information organization (hierarchical vs. network-based)
    - Linearity vs. parallelism of thought
    - Ease vs. difficulty of prioritization
    - Planning-oriented vs. improvisational tendencies
2. **Dominant Modality Scale (Language-Dominant vs. Body-Dominant)**
    - Thinking style (verbal vs. visual/bodily)
    - Learning style (reading/listening vs. seeing/doing)
    - Expressive style (speaking/writing vs. moving/making)
    - Problem-solving style (logical analysis vs. trial-and-error)
3. **Metacognitive Level Scale**
    - Awareness of one's own cognitive processes
    - Ability to verbalize cognitive styles
    - Flexibility in adjusting cognition to context
    - Understanding of personal strengths and limitations
4. **Translation Cost Scale**
    - Cognitive burden (fatigue, concentration difficulties)
    - Temporal cost (slower processing speed)
    - Bodily symptoms (headaches, gastrointestinal issues, sleep disturbances)
    - Emotional cost (stress, self-denial, depressive affect)

Large-Scale Survey Design

- Sample size: 1,000+ participants (with and without diagnoses)
- Estimation of the distribution across the eight quadrants
- Correspondence between existing diagnoses (ASD, ADHD, AuDHD) and the model
- Relationship between translation cost and well-being
- Testing the independence of the three dimensions (factor analysis, structural equation modeling)

Neuroscientific Research: Neural Bases of the Eight Quadrants

Neuroimaging methods such as fMRI, structural MRI, and diffusion tensor imaging (DTI) should be used to explore the neural substrates of the eight-quadrant model.

Key research questions include:

1. **Neural bases of hierarchical vs. non-hierarchical cognition**
    - Activity patterns in regions associated with hierarchical processing (prefrontal cortex, parietal cortex)
    - Network-based processing and default mode network activity
    - Patterns of functional connectivity (hierarchical vs. distributed)
2. **Neural bases of dominant modality**
    - Activation of language regions (Broca's area, Wernicke's area)
    - Activation of motor, somatosensory, and visual cortices
    - Hemispheric dominance (left vs. right)
    - Differences in white matter connectivity (language vs. visuospatial networks)
3. **Neural bases of metacognitive level**
    - Structure and function of the frontal pole
    - Activity in prefrontal control networks
    - Brain activation patterns during metacognitive tasks
4. **Neural correlates of translation cost**
    - Cognitive load during dual processing (non-hierarchical cognition plus hierarchical translation layer)
    - Working memory occupation (prefrontal and parietal regions)
    - Stress responses (amygdala, hypothalamic–pituitary–adrenal axis)

Integration with Existing ASD/ADHD Research

- Whether locally increased connectivity and reduced long-range connectivity reported in ASD reflect hierarchical cognition
- Whether default mode network hyperactivity reported in ADHD reflects non-hierarchical cognition
- Reinterpretation of existing findings through the eight-quadrant framework

6.3.2 Theoretical Refinement

The second direction is further refinement of the theoretical framework presented in this paper.

Detailed Descriptions of the Eight Quadrants

While this paper outlines the eight quadrants, each requires a more detailed account.

For each quadrant, future work should describe:

- Internal cognitive structures (information processing, decision-making, learning styles)
- Manifestations in daily life (school, work, relationships, leisure)
- Strengths and vulnerabilities (contexts of optimal functioning vs. difficulty)
- Typical developmental trajectories
- Optimal forms of support and environments

Deepening the Concept of Dominant Modality

Although the present theory adopts a language-dominant vs. body-dominant distinction, further elaboration is needed.

Key questions include:

- **Modality diversity**: Potential subtypes such as visually dominant or auditorily dominant cognition
- **Plasticity**: Developmental and training-related changes in modality dominance
- **Modality and ASD subtypes**: Correspondence between language-dominant ASD and body-dominant ASD
- **Modality and ADHD subtypes**: Correspondence between inattentive (language-dominant) and hyperactive (body-dominant) presentations
- **Mixed types**: States in which both modalities are simultaneously activated

Developmental Mechanisms of Metacognition

A theoretical account is needed of how metacognitive level is formed and transformed.

Key questions include:

- **Genetic factors**: Heritability of metacognitive capacity
- **Developmental factors**: Critical periods and stages of metacognitive development
- **Environmental factors**:
  - Supportive environments that promote self-understanding
  - Suppressive environments that enforce adaptation
  - Effects of educational interventions targeting metacognition
- **Stabilization at moderate metacognition**: Why AuDHD often remains at a moderate level
- **Transition to high metacognition**: Conditions enabling the emergence of the galaxy type

Structure and Function of the Translation Layer

The translation layer in hybrid types (AuDHD, galaxy type) requires more precise theorization.

Key questions include:

- **Acquisition process**: How translation layers are constructed
- **Qualitative differences** between moderate and high metacognition
- **Stability**: Whether translation layers can deteriorate or be lost
- **Automation**: Transition from effortful to automatic processing
- **Quantification of translation cost**: Methods for measurement and comparison

Mechanisms of Embodiment of Translation Cost

Further theorization is required regarding how translation cost manifests as bodily symptoms.

Key questions include:

- Relationship between cognitive load and bodily symptoms
- Involvement of stress-response systems (HPA axis, autonomic nervous system)
- Prioritization failures in bodily control (e.g., mechanisms underlying incontinence)
- Persistence of chronic symptoms (headaches, gastrointestinal issues, sleep disorders)
- The body as a site of resistance to social protocols

Integration with Other Cognitive Theories

Finally, the theory should be integrated with existing frameworks in cognitive science, developmental psychology, and neuroscience.

Potential connections include:

- **Network and graph theory**: Mathematical modeling of non-hierarchical cognition
- **Complex systems science**: Self-organization and emergence in cognitive systems
- **Connectomics**: Correspondence between brain connectivity and cognitive type
- **Executive function theories**: Differences in executive control across cognitive structures
- **Learning theories**: Connections with Kolb's experiential learning theory (see Appendix)
- **Developmental theories**: Dialogue with Piagetian and Vygotskian models

6.3.3 Applied and Practice-Oriented Research

The third direction is applied research aimed at experimentally evaluating the effects and feasibility of cognitive segregation.

Educational Experiments: Evaluating Exploratory Education

Schools or programs implementing exploratory education optimized for non-hierarchical cognition should be established and evaluated.

Design principles of exploratory education:

- Interest-driven, non-sequential learning
- Project- and portfolio-based assessment instead of examinations
- Exploration and facilitation rather than lecture-based instruction
- Flexible time management instead of fixed schedules
- Multi-modal learning beyond language-centered instruction

Outcome measures:

- Learning outcomes (depth of understanding, creativity)
- Well-being (stress, self-efficacy, school satisfaction)
- Reduction of translation cost (fatigue, bodily symptoms)
- Social skills (collaboration, communication)
- Long-term outcomes (education, employment, life satisfaction)

Comparative designs:

- Exploratory vs. curriculum-based education
- Non-hierarchical vs. hierarchical students
- Longitudinal follow-up

Labor Experiments: Evaluating Project-Based Work

Organizations centered on project-based work optimized for non-hierarchical cognition should be established and studied.

Design principles:

- Flat network structures rather than hierarchical organizations
- Project-based collaboration rather than long-term positional employment
- Flexible schedules instead of fixed working hours

- Autonomous decision-making rather than top-down instruction
- Evaluation based on contribution and creativity rather than promotion

Outcome measures:

- Productivity and innovation
- Job satisfaction, stress, burnout
- Reduction of translation cost (fatigue, turnover)
- Quality of collaboration
- Economic sustainability

Comparative designs:

- Project-based vs. hierarchical organizations
- Non-hierarchical vs. hierarchical workers

Community Experiments: Building Non-Hierarchical Communities

Long-term observational studies should be conducted in communities optimized for non-hierarchical cognition.

Design principles:

- Flexible spatial use rather than fixed zoning
- Project-based activities instead of committees and roles
- Improvisational mutual aid rather than planned welfare
- Network-based decision-making rather than majority rule
- Gift and relational economies coexisting with monetary systems

Outcome measures:

- Well-being (life satisfaction, mental and physical health)
- Social relationships (trust, reciprocity, sense of belonging)
- Reduction of translation cost
- Economic sustainability
- Community stability (retention, conflict resolution)

Comparative designs:

- Non-hierarchical communities vs. hierarchical societies
- Longitudinal follow-up (5–10 years)

6.3.3.1 Ethical Considerations in Applied Research All applied research must be based on voluntary participation. Coercion or harm for experimental purposes is unacceptable. Learning from failure and iterative redesign are essential. Cognitive segregation is not a one-time solution but a process refined through trial and error.

6.4 Social Implications

If the present theory is valid, it calls for a fundamental redesign across multiple domains of society.

6.4.1 Redesigning Diagnosis and Support

Medical and welfare systems must shift from a model centered on *treating disorders* to one focused on *understanding cognitive types and optimizing environments.* Diagnosis should no longer function as a measure of how well an individual adapts to a hierarchical society, but rather as a tool for understanding that individual's cognitive structure. Diagnostic labels such as ASD and ADHD would thus come to signify "you have a different cognitive type," rather than "you have a disorder that must be corrected."

Support should move away from assimilation training into hierarchical norms and toward facilitating access to environments suited to each cognitive type. Pharmacological and behavioral interventions should be positioned not as means of changing the person, but as auxiliary measures that may temporarily reduce translation cost. Genuine support consists in providing segregated spaces: institutional and economic support that enables non-hierarchical individuals to access education, work, and communities optimized for non-hierarchical cognition.

6.4.2 Transformation of the Educational System

Education must transition from a single, curriculum-based system to a multi-track structure. Stepwise, systematic education for hierarchical cognition and exploratory, project-based education for non-hierarchical cognition should coexist, allowing children to choose according to their cognitive type.

This theory also calls for a reexamination of existing learning theories. For example, Kolb's learning style theory proposes four styles, yet contemporary education systems heavily privilege the assimilating style (reflective observation → abstract conceptualization). This bias reflects optimization for hierarchical cognition and fails to capture non-hierarchical learning modes such as parallel exploration and non-linear association. Moreover, Kolb's theory assumes relatively fixed individual learning styles, whereas the present framework suggests that the development of metacognition may enable individuals to flexibly employ multiple styles. Educational design should therefore assume cognitive diversity and support the coexistence of multiple learning modes.

Evaluation systems must also expand beyond hierarchical scores and rankings to include diverse forms of assessment, such as portfolios, project outcomes, and peer review.

6.4.3 Diversification of Work Environments

Work must allow organizational and project-based modes to coexist on equal footing. Hierarchical organizations—companies, formal roles, and long-term employment—are well suited to hierarchical cognitive types. Project-based work—freelancing, collaborative projects, and short-term contracts—is better suited to non-hierarchical cognitive types.

At present, project-based work is often marginalized as precarious employment. In a segregated society, however, it would be recognized as a legitimate and valued mode of work. Social security systems, labor law, and taxation would need to adapt accordingly.

Evaluation systems should move away from hierarchical promotion toward criteria such as contribution to projects, creativity, and collaborative capacity. Working time, likewise, should shift from rigid schedules to flexible time management.

6.4.4 Broader Implications

The implications of this theory extend beyond education and labor. Urban design, legal systems, political institutions, and culture as a whole are heavily biased toward hierarchical cognition. Cognitive segregation suggests the need to redesign all of these domains.

This represents an immense challenge—but also an immense opportunity. By constructing a society in which each cognitive type can function optimally, humanity's overall capacity and creativity can be liberated. Hierarchical cognition excels in system-building and knowledge accumulation; non-hierarchical cognition excels in exploration and creation. Cognitive segregation is a form of social design that seeks to fully realize this diversity.

7 Conclusion

This paper has reinterpreted developmental disorders as differences in cognitive structure, elucidated the translation costs imposed on non-hierarchical cognition by a hierarchical society, and proposed *segregation* as a new model of social design. In this concluding chapter, the core arguments of the paper are summarized, the significance of segregation is reaffirmed, and future prospects are outlined.

7.1 Summary of the Hypothesis

The central claim of this paper is that human cognition consists of two fundamentally different types: **hierarchical cognition** and **non-hierarchical cognition**. In hierarchical cognition, information is organized in hierarchical or tree-like structures; in non-hierarchical cognition, information exists in network- or graph-like structures. This distinction is not merely a difference in "style," but a fundamental difference in how the world is experienced.

Existing diagnoses of developmental disorders can be reinterpreted as manifestations of these differences in cognitive structure. **Autism Spectrum Disorder (ASD)** reflects the difficulty hierarchical cognition encounters when faced with elements that resist hierarchization, such as implicit social rules, fluid relationships, and the sensory world. **Attention-Deficit/Hyperactivity Disorder (ADHD)** reflects the difficulty non-hierarchical cognition faces in adapting to a hierarchical society structured around planning, prioritization, and fixed rules. Many cases of comorbid ASD and ADHD represent a **hybrid type**, in which a hierarchical layer is acquired secondarily on top of a non-hierarchical cognitive base.

Modern society is optimized for hierarchical cognition at every level—language, institutions, economy, education, and medicine. Individuals with non-hierarchical cognition are compelled to bear a chronic burden of "translating" their cognitive style into a hierarchical one. This **translation cost** accumulates as cognitive, temporal, and emotional load, manifesting as physical symptoms, mental exhaustion, and social maladaptation. Crucially, this translation cost is borne unilaterally by non-hierarchical individuals, revealing a profound structural asymmetry.

7.2 The Significance of Segregation

To address this problem, this paper proposes not "coexistence," but **segregation**. Segregation refers to a social arrangement in which each cognitive type lives in environments optimized for its own cognitive structure. This segregation is realized spatially, institutionally, and informationally. Hierarchical and non-hierarchical societies coexist in parallel, and individuals choose between them according to their cognitive type. Boundaries are clear, but interaction is free, mediated by translators.

The significance of segregation is threefold.

First, it enables liberation from translation costs. Non-hierarchical individuals are no longer forced to translate themselves into hierarchical forms. They can live naturally, retaining their own cognitive mode. This represents the restoration of **cognitive dignity**.

Second, segregation allows for the maximization of each cognitive type's strengths. Hierarchical cognition excels in system construction, knowledge accumulation, and institutional design; non-hierarchical cognition excels in exploration, creativity, and adaptation. However, these capacities can flourish only in appropriate environments. By enabling each type to live in its optimal environment, segregation unlocks the full potential and creativity of humanity as a whole.

Third, segregation enables genuine diversity. Forced coexistence under the name of inclusion is, in practice, pressure toward assimilation into hierarchical norms. Segregation allows each cognitive type to exist on equal footing and to develop its own culture. Diversity does not mean absorption into uniformity; it means respect through differentiation.

7.3 Future Prospects

This paper presents a theoretical proposal. In the future, it must be tested, refined, or rejected through empirical research. Elucidating neural substrates, developing measurement tools, and conducting large-scale surveys are all necessary. The validity of this theory will be determined through such investigations.

At the same time, practical experiments in segregation are required. Small-scale implementations of exploratory education, project-based labor, and non-hierarchical communities can be established to assess the effects of segregation. Both successes and failures will provide valuable insights. Through incremental experimentation, the feasibility of segregation can be progressively enhanced.

In the long term, a comprehensive redesign of society comes into view. Education, labor, the economy, cities, and legal systems—all can be restructured on the assumption of cognitive diversity. This is a formidable challenge, but not an impossible one. Humanity has repeatedly transformed its social structures in the past. The Industrial Revolution, the expansion of democracy, and the establishment of human rights all once seemed impossible, yet were realized. Segregation, likewise, is achievable.

7.4 Final Message

To those directly affected who read this paper: the vague sense of discomfort you have felt has a name. It does not mean you are "broken," but that you are a different type. The translation imposed on you by a hierarchical society is a form of violence. You have the right to live without translation. Demand segregation.

To members of hierarchical society who read this paper: your society is not "standard," but merely one type among others. The imposition of translation costs on non-hierarchical individuals is a form of structural violence. Accept segregation. It will be a form of liberation for you as well.

To researchers who read this paper: test this theory. Criticize it. Improve it. Or reject it. Science advances through competition among theories, and this theory, too, should be subjected to that process.

Cognitive diversity is one of humanity's strengths. But for that diversity to be genuinely respected, segregation is necessary. Beyond assimilation under the name of coexistence, toward a new form of shared existence grounded in segregation. That is the proposal of this paper.

8 Acknowledgments

This paper was shaped through many years of dialogue with myself, as well as through sustained dialogue with others.

Parallel and iterative dialogues with multiple major large language models (LLMs) played a decisive role in the conception and writing of this paper. These AIs did not function merely as writing aids; rather, they served as **"cognitive mirrors" and "translation engines"** that translated the author's non-hierarchical (galactic-type) cognition into hierarchical language. In particular, the very process of selectively engaging with different AI models according to their distinct cognitive characteristics—such as tolerance for fluid, exploratory thinking versus guarantees of rigorous structural organization—made it possible to theorize the concept of **"translation cost,"** which lies at the core of this paper. Without this multi-perspectival dialogue, this paper would not have come into existence.

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Finally, to all those who live with a similar, unnamed sense of dissonance: I hope that this paper gives words to that feeling and serves, even in a small way, as a light pointing toward the path of *segregation*.

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10 Appendix

10.1 Supplementary Material: Critique of Kolb's Learning Theory

Reinterpretation from the Two-Type Theory of Human Cognition

10.1.1 Introduction

This supplementary material attempts to apply the Two-Type Theory of Human Cognition (hierarchical/non-hierarchical) presented in the main paper to the field of learning theory. Specifically, it critically examines David Kolb's Experiential Learning Theory and his learning style model, reinterpreting them from the perspective of cognitive types.

Kolb's learning theory is widely influential in educational practice. However, from the perspective of the present theory, Kolb's theory has the following issues: 1. The educational system is heavily biased toward the assimilating style, which is optimized for hierarchical cognition. 2. It overlooks the potential for high metacognition to transcend specific learning styles. 3. It fails to adequately capture non-hierarchical learning styles (parallel, jump-wise).

This material elaborates on these issues and suggests the possibility of a new learning theory based on cognitive types.

10.1.2 Overview of Kolb's Learning Theory

10.1.2.1 The Experiential Learning Cycle Kolb (1984) formalized learning as a four-stage cycle:

1. **Concrete Experience (CE)**: Actual hands-on experience.
2. **Reflective Observation (RO)**: Reflecting on the experience.
3. **Abstract Conceptualization (AC)**: Forming theories and concepts.
4. **Active Experimentation (AE)**: Testing in new situations.

It is believed that learning deepens by rotating through this cycle.

10.1.2.2 The Four Learning Styles Kolb proposed four learning styles based on which stages of the cycle an individual prefers:

- **Diverging**: CE + RO. Emphasizes diverse perspectives, imagination, and emotion.
- **Assimilating**: RO + AC. Emphasizes logical thinking and abstract models.
- **Converging**: AC + AE. Emphasizes problem-solving and practical application.
- **Accommodating**: AE + CE. Emphasizes intuition, trial-and-error, and adaptation.

Individuals are generally thought to possess one primary style, and education is advised to align with this style.

10.1.3 Critique 1: Bias toward the Assimilating Style in Social Systems

10.1.3.1 Structure of the Educational System In reality, educational systems are heavily biased toward the **Assimilating style** (RO + AC: Reflection → Abstract Conceptualization).

Typical lesson flow: 1. Teacher lectures (presentation of abstract concepts). 2. Students reflect and comprehend (Reflective Observation). 3. Exams evaluate abstract understanding (confirmation of Abstract Conceptualization).

Concrete Experience (CE) and Active Experimentation (AE) are limited to specific settings like labs or field work. The mainstream of education is the sitting transmission of abstract knowledge.

10.1.3.2 Correspondence Between Assimilating Style and Hierarchical Cognition Assimilating learning (Reflection → Abstract Conceptualization) corresponds directly to hierarchical cognition.

Assimilating learning process: * Analyzing experiences reflectively. * Extracting common patterns (abstraction). * Systematizing abstract concepts hierarchically. * Explaining subordinate instances from superordinate concepts.

This is precisely the characteristic of hierarchical cognition, which organizes information hierarchically. Hierarchical individuals naturally excel at this process.

10.1.3.3 Difficulty for Non-Hierarchical Individuals

For non-hierarchical learners, the assimilating style is unnatural.

Non-hierarchical learning style: * Associating experiences in a network fashion. * Retaining multiple instances in parallel. * Jumping associatively between different domains. * Emphasizing concrete association over abstraction.

The current educational system negatively evaluates this learning style as "shallow" or "unsystematic." However, the problem lies not with the learner, but with the system's bias toward hierarchical cognition.

10.1.4 Critique 2: Overlooking Metacognitive Transcending of Styles

10.1.4.1 Assumption of Fixed Styles

Kolb's theory assumes that individuals possess relatively fixed learning styles. Education is expected to accommodate these fixed styles.

However, this assumption is problematic. With the development of metacognition, individuals can utilize multiple learning styles flexibly.

10.1.4.2 Metacognition and Learning Flexibility

Metacognition is the ability to recognize and regulate one's own cognitive processes. Learners with high metacognition can: * Understand their own natural learning style. * Select a different style depending on the situation. * Consciously utilize even non-preferred styles.

For example, when a non-hierarchical learner acquires metacognition: * They recognize their own parallel, jump-wise learning style. * They can consciously "**translate**" their learning into the assimilating style required by a hierarchical educational system. * They write hierarchical answers on exams while maintaining their non-hierarchical learning approach privately.

This is the same mechanism described in the main paper for the formation of the **hybrid type**.

10.1.4.3 Kolb's Oversight

Kolb's theory overlooks this flexibility enabled by metacognition. Learning styles are not fixed but developmental and context-dependent. The goal of education should not be to accommodate a specific style, but to foster metacognition so that learners can master and utilize multiple styles.

10.1.5 Critique 3: Invisibility of Non-Hierarchical Learning Styles

10.1.5.1 Hierarchy in the Kolb Cycle

Kolb's four-stage cycle implicitly assumes a hierarchical progression.

Implicit assumptions of the cycle: * The order: Experience → Reflection → Conceptualization → Experiment is necessary. * Each stage must be "stepped through." * Skipping stages or reverse progression constitutes "incomplete learning."

However, this is a hierarchical model of learning. Non-hierarchical learning does not conform to this cycle.

10.1.5.2 Non-Hierarchical Learning: Parallel and Jump-Wise

Non-hierarchical learners: * Engage in Experience, Reflection, Conceptualization, and Experimentation **simultaneously and in parallel**. * Jump directly from one experience to another (without sequential reflection or conceptualization). * Understand through a **network of concrete examples** rather than abstract concept formation.

For example, in programming: * Hierarchical type: First learns theory (data structures, algorithms), then implements. * Non-hierarchical type: Explores multiple concrete examples while implementing, and discovers patterns associatively.

While both are valid forms of learning, Kolb's theory implicitly assumes only the former as "correct learning."

10.1.5.3 Confusion of Diverging and Non-Hierarchical Kolb's "Diverging" style superficially appears similar to non-hierarchical cognition, emphasizing diverse perspectives and associative thinking. However, the Diverging style remains within the framework of the four-stage cycle. True non-hierarchical learning transcends the cycle itself.

10.1.6 Restructuring Learning Theory Based on Cognitive Types

10.1.6.1 Characteristics of Hierarchical Learning **Information processing:** * Systematic and sequential. * Constructs a hierarchy from abstract $\rightarrow$ concrete. * Learns theory first, then applies.

Suitable learning environment: * Lecture format. * Systematic learning via textbooks. * Explicit evaluation criteria (exams, grades).

Suitable domains: * Mathematics, physics, law, and other domains where systematicity is crucial.

10.1.6.2 Characteristics of Non-Hierarchical Learning **Information processing:** * Exploratory and parallel. * Constructs a network of concrete examples. * Discovery through trial-and-error.

Suitable learning environment: * Project-based learning. * Exploratory self-learning. * Portfolio assessment.

Suitable domains: * Art, design, entrepreneurship, and other domains where creativity is crucial.

10.1.6.3 Potential of Hybrid Learning Hybrid learners with developed metacognition: * Utilize both hierarchical and non-hierarchical styles depending on the situation. * Use the hierarchical style for theoretical learning, and the non-hierarchical style for creative activities. * Can integrate both approaches as "translators" of learning.

The ideal of education is to foster this hybrid type. However, the current system trains only the hierarchical style, compelling non-hierarchical learners to pay continuous translation costs.

10.1.7 Implications for Educational Practice

10.1.7.1 Necessity of Multiple Tracks Instead of a single educational system (optimized for hierarchical cognition), multiple tracks are necessary:

Hierarchical Track: * Curriculum-based. * Sequential and systematic. * Lectures and examinations.

Non-Hierarchical Track: * Project-based. * Exploratory and jump-wise. * Portfolios and self-assessment.

Students choose based on their cognitive type.

10.1.7.2 Importance of Metacognition Education Metacognition development is crucial in both tracks: * Understanding one's own learning style. * Experiencing different learning styles. * Flexible utilization of styles based on context.

Metacognition enables the "transcending of types" discussed in the main paper.

10.1.7.3 Changing Role of the Teacher

Teachers must utilize multiple teaching methods rather than a single pedagogy: * Systematic lectures for hierarchical students. * Exploratory facilitation for non-hierarchical students. * Understanding student cognitive types and providing appropriate learning environments.

10.1.8 Conclusion

Kolb's learning theory has made significant contributions to educational practice. However, it suffers from a bias toward hierarchical cognition, the neglect of metacognition, and the invisibility of non-hierarchical learning.

From the perspective of the Two-Type Theory of Human Cognition, learning theory must also be restructured on the premise of cognitive diversity. Recognizing hierarchical and non-hierarchical learning styles as equally valid, offering multiple tracks, and fostering the ability to master both through metacognition—this is the vision of an education that respects cognitive diversity.

This supplementary material has offered merely a glimpse of the implications for learning theory. Further detailed theorization and empirical research are required. Education is one of the most crucial domains of social systems and plays a central role in realizing cognitive segregation.

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This supplementary material is an appendix to the main paper and is not a standalone article. It provides a more detailed development of the critique of Kolb's learning theory. In the future, it may be possible to develop this into an independent paper based on this foundation.